A Project Stage-II
Report on

Determination of Relationship between implant stability quotient and resonance frequency analysis

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C E R T I F I C A T E

This is to certify that ***Mr.Mihir Adhikari, Mr. Aditya S. Badgujar, Mr. Shreyas R. Rajebahadur and Mr. Kanad Dhok*** has successfully completed the Project Stage - II entitled ***"Determination of Relationship between implant stability quotient and resonance frequency analysis"*** under my supervision, in the partial fulfilment of Bachelor of Technology - Mechanical Engineering of Dr. Vishwanath Karad MIT World Peace University, Pune.

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It gives me great pleasure to present a mini project on **"Determination of Relationship between implant stability quotient and resonance frequency analysis".** In presenting this mini project work, a number of hands helped me directly and indirectly.

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Nomenclature

DI – Dental Implant

ISQ – Implant Stability Quotient

RFA – Resonance Frequency Analysis

FEA - Finite Element Analysis

RF – Resonance Frequency

OMI – Orthodontic Mini Implant

Abstract

There are many factors affecting the success of dental implants. One of the most important factors is the implant stability. At the time of implantation, some amount of bone is removed during the surgical procedure. Due to this, there exists micro-gap between the implant and the surrounding bone. Over a period of time, the bone fuses with the implant due to osseointegration. Hence, before complete osseointegration is achieved, it is essential to measure the primary stability of the implant to gauge its displacement that may occur in the micro-gap. Depending on factors as the patient's bone density, implant design and loading, the rate of osseointegration varies. Once the primary stability is measured, it gives the dentist an opportunity to make an informed decision regarding the loading protocol and post-operative treatment plants. Thus, the current project aims to develop relation between Resonance Frequency Analysis and Implant Stability Quotient to accurately measure the stability of the implant.

Keywords: Dental Implants, Finite Element Analysis, Implant Stability Quotient, Resonance Frequency Analysis.

Chapter-1

Introduction

Stability of an implant is a crucial factor in osseointegration. As osseointegration, which is the bone remodeling and regeneration occurring at implant surface is directly dependent on primary stability, it is extremely important to accurately measure the primary stability not only at the time of implant placement but also at various stages of healing. It is, therefore, critical to use a method of measurement such that it continuously monitors implant stability at regular intervals of time during the healing process. Over several decades, numerous methods have been employed to find implant stability. These methods are 8categorized as invasive and non-invasive. Historically, destructive methods such as tensional test and removal torque test were used extensively to measure implant stability. However, they have lost credibility due to their invasiveness and ethical issues. In recent times, non-invasive techniques such as RFA (Resonance Frequency Analysis), reverse torque analysis, have gained popularity. This project explains the RFA technique to measure dental implant stability quotient and discusses the relationship between both of them.

1.1 Dental Implants:

A dental implant is a surgical fixture that is placed into the jawbone and allowed to fuse with the bone over the span of a few months. The dental implant acts as a replacement for the root of a missing tooth. Most dental implants are made of titanium, which allows them to integrate with bone without being recognized as a foreign object in our body. Due to advancement in medical field and technology, today the success rate for dental implants is quite high. While the primary function of dental implants is for teeth replacement, there are areas in which implants can assist in other dental procedures. Due to their stability, dental implants can be used to support a removable denture and provide a more secure and comfortable fit. A dental implant consists of 3 components which are Fixture (implant), Abutment and Crown.

(1) Fixture (Implant): It is a small, screw-shaped structure placed under the gums. A surgeon drills a small hole where the missing tooth’s roots once were, the fixture is surgically placed in the jawbone, acting as the implant’s “roots.” During the healing process, the fixture fuses with the jawbone in a process known as osseointegration, which allows the implant to reside in your mouth permanently. Usually made with titanium [1].

(2) Crown: is the part of a dental implant that looks and functions exactly like a real tooth. It can be made from a variety of materials such as porcelain or ceramic, and it can either be screwed into place or cemented on the abutment.

(3) Abutment: is a short, thick screw that extends at or directly above the gumline to support the tooth replacement. A separate screw is used to connect the crown with abutment. Abutment acts as a connector between the fixture inside the gumline and the replacement tooth (crown). It can either be made from metal or a tooth-coloured material and can either be separate from the fixture or a single integrated unit. A separate abutment is typically not attached until after osseointegration has occurred so that the fixture has enough time to heal and settle.

Figure 1: Dental Implant Components

1.2 Osseointegration:

Osteointegration is the bonding between the implant metal and cancellous bone cells in the human body. Over a period of time after the dental surgery, the dental implant fuses with the surrounding bone and osseointegration takes place. To ensure a long-term success of these implants, the osseointegration process must be acceptable enough to hold the implant firmly into its position. The process of osseointegration is largely influenced by primary stability of the implant and regeneration of the bone. Hence, osseointegration determines the dental implant stability [2].

1.3 Implant Failure:

Despite the high success rate of dental implants, they do fail. A lack of primary stability, surgical trauma, and infection seem to be the most important causes of early implant failure. Implant failure is a static outcome situation that requires removal of a failed implant. Some of the factors contributing to the early and late Implant failure are [3]:

- Poor bone quality and quantity
- Bone healing and general health condition
- Clinical infection

- Post-insertion pain due to pressure acting on soft tissue
- Lack of primary stability due to improper osseointegration.
- Inadequate surgical and prosthetic techniques
- Excessive loading and Inadequate prosthetic construction

1.4 Implant Stability:

The assessment of dental implant stability is crucial for dentists to complete the operation successfully and reduce the probability of treatment failure and potential medical disputes. Poor osseointegration between the implant and alveolar bone causes post-implant instability, and consequently leads to surgery failure [4]. Post-implant stability can be divided into primary stability and secondary stability. Early works on implant stability categorized the implant stability in two stages as primary and secondary stability. The majority of implant losses occur due to biomechanically induced failures, caused because of low primary implant stability, low bone density, short implants and overload. Hence, achievement and maintenance of firm implant stability are regarded as preconditions for a successful clinical outcome with dental implants [4], [5].

(i) Primary Stability:

The primary stability is referred to as biomechanical compatibility which starts immediately after the implantation. The primary stability is affected by several factors such as bone quantity and quality, the exterior design of an implant including geometric shape, length, diameter, and surface treatment, implant design, patient’s bone conditions, and technique and protocol followed by physician during surgery. Measurement of the primary stability is of utmost importance to estimate about the success of an implant fixation by evaluating the sufficient level of osseointegration and hence prevent implant failure [6].

(ii) Secondary Stability:

While primary stability is related to the mechanical relationship between the implant and the bone, secondary stability, also referred to as biological stability is related to bone regeneration and remodelling after implantation. Secondary stability is achieved with time as a result of bone healing, where, newly formed bone will bridge and fill the voids of the interface zone and grow into surface irregularities and macroscopic undercuts, which results in interlocking and further stabilization of the implant. The newly formed bone matures with time, which results in an increased density and stiffness of the implant-bone complex. Changes in implant stability may depend on the degree of osseointegration, which is affected by various factors during the healing period. The evaluation of implant stability at various times may provide significant information as to the individualized optimal healing time [6].

From the literature it is understood that the primary stability is associated with the mechanical engagement of an implant with the surrounding bone and is of utmost importance for implant's stability. Because of such importance various methods are developed to measure the Primary Stability since, it is a mechanically dependent parameter which include use of Radiographic Analysis, Reverse Torque Technique (RTT), Periotest, Impact hammer test, Resonance Frequency Analysis (RFA), Finite Element Analysis (FEA) and Ultrasound Method are a few of the widespread methods [4]. Among these methods, RFA has been a popular choice for several scientific studies on implant stability.

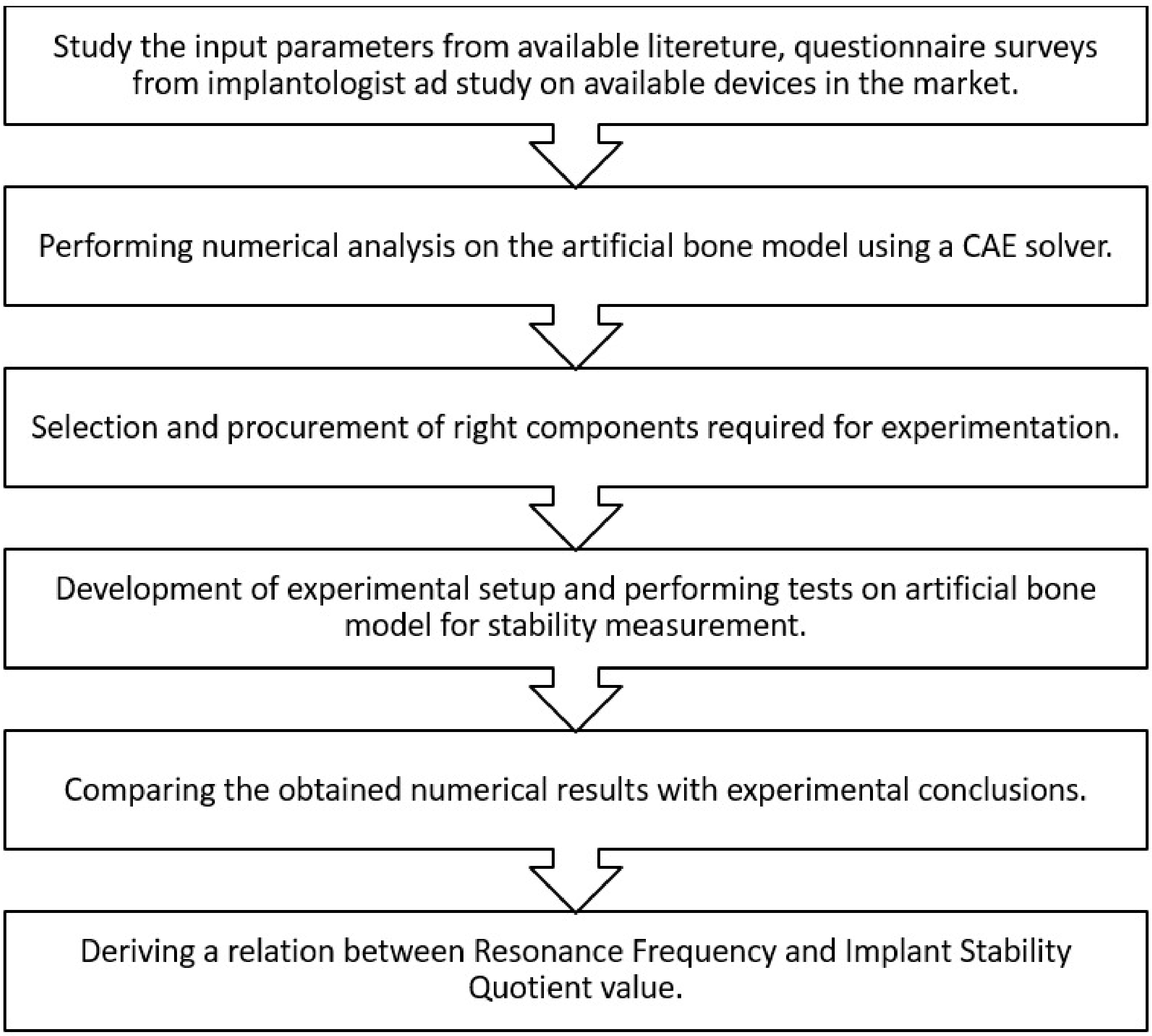

Figure. 2 Step by step methodology of the project [7]

1.5 Resonance Frequency Analysis (RFA):

The RFA technique is based on the vibration analysis of dental implants under external excitation which may be of contact (invasive) or non-contact type (non-invasive). In the recent two decades the RFA method has been explored and become a main-stream detection method [4]. Electromagnetic RFA provides a non-invasive, objective method of assessing implant stability over time. This technology has been studied and utilized in implant dentistry for more than 20 years and can be defined as the measurement of the frequency at which an object vibrates. As the stability of an implant increases, the frequency at which it will vibrate increases as well; the increase in frequency reflects increased stiffness of the bone-to-implant interface, which is thought to reflect osseointegration. A lesser lateral movement of the implant due to the vibrations has been associated with higher RFA values. In addition, higher bone density has been positively correlated with higher RFA values. This method can potentially provide clinically relevant information about the state of the interface between the implant and the bone at any stage of the treatment [8] [9].

1.6 Working Principle of Electromagnetic RFA:

RFA works on the principle that when a dental implant is repeatedly vibrated by a frequency in the audible range, resonance occurs at a higher frequency as the strength of the bone- implant interface increases.

RFA uses a transducer to apply a small magnitude bending force to the implant-bone system. This bending force mimics the loading condition clinically by applying a constant force in the lateral direction of the implant and measuring the resulting displacement. The resonance frequency is affected by the stiffness at the bone-implant interface and the effective length of the fixture due to bone loss. Implant stability is evaluated as a function of the stiffness at the bone-implant interface. As a result, any change in the stiffness value is assumed to represent a change in the level of osseointegration at the bone-implant interface. Thus, as the bonding level of the implant with the surrounding bone/tissue complex increases, the resonant frequency will also increase. Eventually, the resonant frequency will attain a constant value, indicating the implant stability after osseointegration. The frequency resulting into maximum amplitude of vibrational displacement are recorded as the resonance frequency [8].

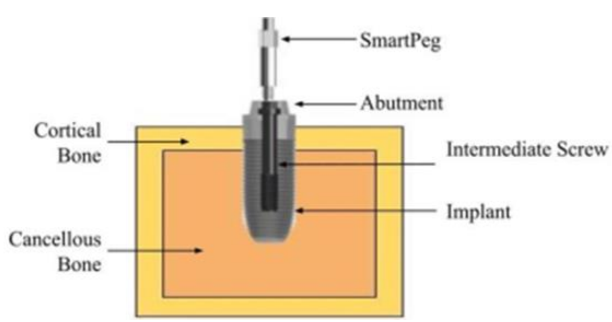

Figure 3: The dental implant system consisting of the implant, SmartPeg and bone block specifications [10]

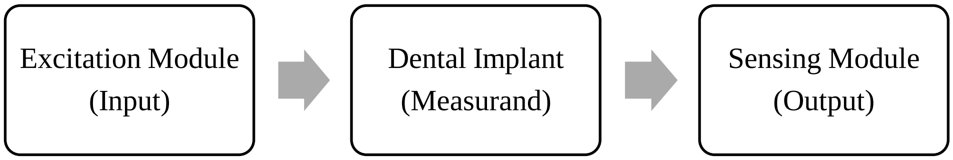

Figure 4: Block diagram of Resonance Frequency Analysis for Implant stability Measurement

Figure 5: (a) Electronic Probe with a ISQ scale monitor, (b) RFA using a SmartPeg sensor mounted on abutment subjected to vibrations induced by the probe

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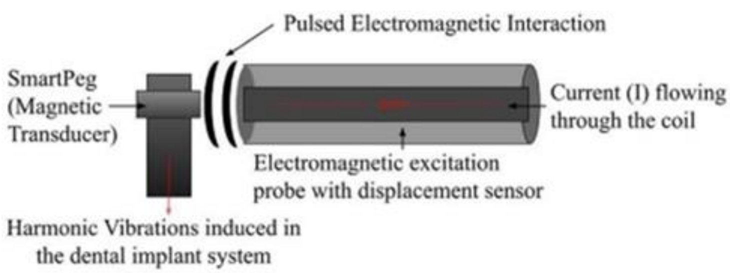


Figure 6: Schematic representation of electromagnetically excited SmartPeg and bone-implant system. [11]

The RFA technique essentially measures the micro mobility of the SmartPeg sensor (and thus the implant) when subject to lateral load. Measuring lateral load is thought to mimic the functional forces that will be placed on the implant once restored.

1.7 Implant Stability Quotient (ISQ):

The resonance frequency obtained is converted into an implant stability quotient (ISQ). This value is mathematically derived from the frequency measurements, and ISQ is developed to create a scale of values, ranging from 1 to 100, that can be tracked and translated into micro mobility and implant stability [4]. Tracking ISQ values throughout the healing process will often demonstrate a decrease in stability over the first three weeks following implant placement. Clinical research has demonstrated the lowest values and most significant reductions in stability occur at sites with poor bone quality. Any observed decrease in stability is thought to be related to bone healing following surgical injury. Understanding the healing process and its effect on implant stability can assist in clinical decision-making related to the timing of implant restoration. In this regard, RFA can provide an objective assessment of implant stability, bone healing and osseointegration [12].

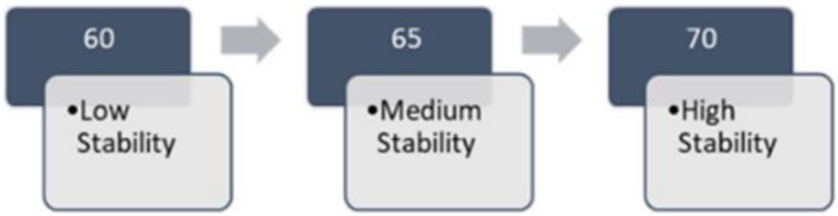


Figure 7: Ostell^{TM} ISQ values corresponding to degree of implant

Although the ISQ scale ranges from 1 to 100, implant ISQ values generally range between 50 and 80. When translating ISQ values into implant stability, it is important to note ISQ values share a nonlinear, inverse relationship with implant micromotion. For example, an increase of ISQ from 60 to 70 reflects a 50% reduction of implant micro mobility. Looking at certain benchmarks for ISQ and implant stability, the following guidelines are often employed: ISQ ≥ 70 = high stability; ISQ 60–69 = medium stability; ISQ < 60 = low stability (Figure 5). Rather than considering a single ISQ value, however, changes or trends in ISQ over time can provide valuable information regarding development of secondary implant stability or osseointegration. Successful implants tend to demonstrate increasing ISQ values during healing, particularly during the first six weeks following surgical placement. Implants with initially low (< 60) or medium (60–69) ISQ levels often demonstrate increases over time, while implants with initially high (≥ 70) ISQ values may remain stable or increase over time.

Chapter-2

Literature Review

Handheld device for assessing dental implant osseointegration through the vibro-acoustic technique (2021)

M.C.Pan et al [1] designed a handheld assessment device which was based on Resonance Frequency Analysis and employed the vibroacoustic technique with acoustic excitation and displacement response sensing to quantify the stability of dental implants. The response spectra calculated described a trend of osseointegration. This method was superior to other methods of dental implant stability measurement because it completely eliminates the need of any magnetic accessory (Smart-Peg). This research consisted of in vitro as well as in vivo testing. In the in vitro testing, the osseointegration was mimicked by using different mixing ratios of resin to hardener epoxy adhesive. Validation was done by doing regression analysis between vibroacoustic technique and Osstell ISQ for in vivo testing and for in vitro testing, statistical analysis was done on measured resonance frequency.

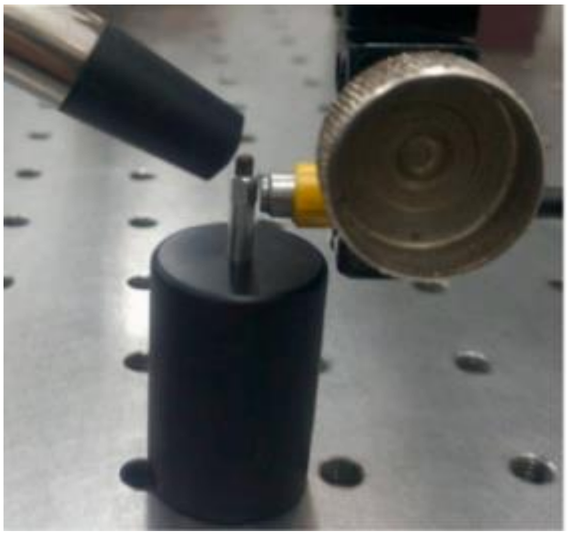


Figure.8 Experimental Setup for determining the stability on dental implant

A Mathematical Model for Biomechanical Evaluation of Micro-motion in Dental Prosthetics using Vibroacoustic RFA

N. Karnik et al (2021) [3] developed a mathematical model for the vibroacoustic method for resonance frequency analysis of dental implants. In this method, the dental implant is excited using either a loudspeaker or a buzzer and the subsequent displacement is captured. The mathematical model is of single degree of freedom which is excited by using a sinusoidal force. The input frequency was varied between 1-50 Hz. The resonance frequency values which corresponded to micromotion values of 1.2 - 2 micrometers were plotted using MATLAB. The average resonance frequency values were 8985 Hz. Notably, the micromotion increased by as much as 40.62 % when the input frequency went above 40 Hz. As a result, it was concluded that higher resonance frequency values indicated a higher dental implant stability.

A finite element model for evaluating the effectiveness of the Advanced System for Implant Stability Testing (ASIST) (2021)

M. Mohamed et al (2021) [4] developed the Advanced system for Implant Stability Testing which predicts the stiffness of bone implant interface. The design's objective is to omit the effects of system geometry depending on the behavior of implant system. Abaqus was used to generate models of implant, abutment, screw, base support and impact rod respectively. Further Mathematica was used to obtain assist stability quotient.

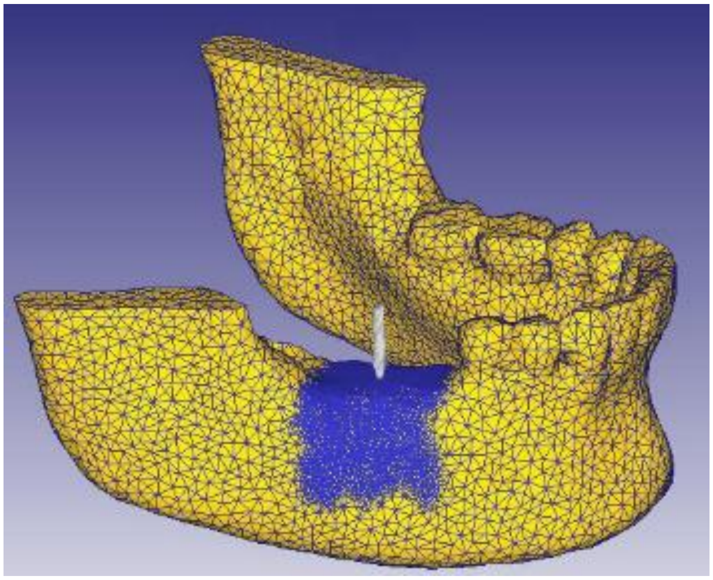


Figure.9 Finite Element Analysis for implant stability testing

Design/exploration and verification of an electromagnetic probe for assessing dental implant osseointegration (2021)

M.C. Pan et al. [5] designed an electromagnetic probe for carrying out in vitro and in vivo tests to assess the osseointegration in dental implants. The titanium implants were put under experimentation by, generating a magnetic pole with the help of dual inductors. The device pulsated 200-10000 Hz of AC current in order to excite the implant. The relationship between RFA and ISQ were found to be close.

Dental implant stability and its measurements to improve osseointegration at the bone-implant interface: A review

N. Kittur et al (2020) [6] analysed all the available methods to assess the dental implant stability, analysed the advantages and disadvantages of each in order to decide the most convenient method to measure the dental implant stability. In this study, all the destructive as well as non-destructive methods were assessed. Destructive methods were, however, ruled out as they were deemed to be unsuitable for long term usage. Resonance Frequency Analysis was found to be the most suitable method for the long-term analysis of dental implants.

Comparison of Measurements of Implant Stability by Two Different Radio Frequency Analysis Systems: An In Vitro Study (2020)

Cenker Zeiki et al. [7] performed this study to compare implant stability by two RFA devices and evaluate them based on ISQ value. They were tested on 20 implants. The average ISQ value of Osstell was 77.60±2.11and that of Penguin was 78.5±2.04. The ISQ values obtained from the Osstell and Penguin machines were significantly compatible ($P < .05$). Both of the RFA devices

Resonant frequency analysis of dental implants

D. Rittel et al (2019) [5] investigated the resonant frequency analysis by the means of a numerical model. In this model, eigenfrequencies were determined and steady state response analysis was done using a commercial finite element package. The sensitivity of resonant frequency analysis to the changes of mechanical properties of the periprosthetic bone tissue was found to be weak. The output amplitude as well as the resonance frequency was more adapted to the estimations of bone healing.

Miniaturized Electromagnetic Device Abutment Improves Stability of the Dental Implants (2018)

S. Barak et al. [8] aim was to study the effects of the miniaturized electromagnetic device (MED) on the implant's stability. They worked on around 28 implants and found out the pulsated electromagnetic field decreases the overall time required for osseointegration. The miniature electromagnetic device abutment conveys a greater stability during the early phase of healing as compared to standard implants.

Application of the advanced system for implant stability testing (ASIST) to natural teeth for noninvasive evaluation of the tooth root interface

L.Westover et al (2018) [9] presents the development of a device that uses the impact method for measurement and analytical model of implant system and its surrounding space to evaluate the stiffness of the interface. The study shows that this technique can be used on the natural teeth to determine the stiffness aspects of the periodontal ligaments.

Non-contact vibro-acoustic detection technique for dental osseointegration examination

H.B Zhuang et al (2014) [10] describes a non-contact type implant stability detecting technique that will use a loudspeaker to to excite and capacitive displacement sensor to determine the stability and the finding of the imperfect locations around the interface of bone and the implant using Resonance Frequency Analysis. In vitro and in vivo studies were performed to justify the efficacy. The obtained results showed that the Resonant frequency differences can be determined and they can be used to evaluate the osseointegration and also the bone defects.

Identifiable range of osseointegration of dental implants through resonance frequency analysis (2010)

S. Wang et al. [11] determines the identifiable range for stiffness of interfacial tissue of dental implants. Dedicated analysis is done in order to reveal the relation between resonance frequencies and certain parameters like, geometry, boundary constrain, material property and interfacial tissue.

Resonance frequency analysis and damping capacity assessment Part 1: an in vitro study on measurement reliability and a method of comparison in the determination of primary dental implant stability (2006)

S. Lachmann et al. [12] found strong correlation between RFA and damping capacity (Periotest) in an in vitro study. The aims of this in vitro study were to evaluate reliability of the Osstell and Periotest devices in the assessment of implant stability and to perform a method comparison. They found Periotest instrument showing greater error in clinical application than in vitro experiments. Similar results were shown in another in vitro study by Lachmann et al. in another work. Results showed agreement between measured implant stability and actual loss of peri-implant resin. They found Osstell instrument to be more precise than Periotest device.

Resonance frequency analysis of one-stage dental implant stability during the osseointegration period (2005)

Ersanli et al. [13] research work was one of the main contributions in determining the implant stability using RFA. They did a study on 122 implants in 31 patients revealed weakest implant stability at 3-6 weeks after placement. They stated that, it is difficult to define a general standardized range of ISQ readings for successful implant integration for various implant systems. Thus, RFA values/ISQ levels should be calibrated for each implant system separately It was concluded that implant stability quotient value can be used to determine different healing phases and the stability of dental implants but implant stability quotient level should be calibrated for each implant system separately. It appears that implant stability is weakest at 3 to 6 weeks in one-stage non-loaded dental implants

Stability measurements of 1-stage implants in the maxilla by means of resonance frequency analysis: a pilot study (2005)

Zix et al. [14] did study with 120 maxillary ITI implants in 35 patients and found mean implant stability quotient was 52.5±7.9. Men showed higher implant stability than women. They concluded that single RFA measurement of an implant does not allow assessment of its current status. Repeated measurements over a longer period of time would be necessary. Also, no significant differences in ISQ values were found between implants with regard to loading period or location in the jaw. Postmenopausal women exhibited significantly lower ISQ values compared to men of the same age group.

Early detection of implant healing process using resonance frequency analysis. (2003)

Huang et al. [15] used natural frequency to assess implant bone interface and they found that the boundary status of an implant can be monitored by detecting its natural frequency. In other in vivo and in vitro studies on implants, they found RFA as a reliable and accurate method for early assessment of osseointegration process. They concluded that these values can not only be used as an indicator for early diagnosis of primary stability but can also provide useful message regarding the secondary stability of implant.

Analysis of the misfit of dental implant-supported prostheses made with three manufacturing processes.

Daniel Delgado et al (2002) [16] presents the development of an instrument which is capable of measuring and noting the information to identify the process of osseointegration in dental implant systems. The parameters identified in this paper that influence the stability are stiffness of the implant components, damping properties of the interface of the implant-bone and also the tissues that surround the interface. The resonance frequency Analysis was done with torsional, longitudinal and transverse vibrations.

A comparison between cutting torque and resonance frequency measurements of maxillary implants. A 20-month clinical study (1999)

Friberg et al. [17] found highest correlation when comparing the mean torque values of the upper cortical portion with the RFA measurements at implant placement. They proved that the stability of implants placed in softer bone seems to get better over time with more dense bone sites. RFA technique proved to be more sensitive in detecting changes of implant stability than the conventional clinical and radiographic examination techniques during other clinical study. One implant failed in this study and lower resonance frequency value indicated the failure several weeks before the mobility was clinically diagnosed.

Chapter-3

Objectives, Problem Statement and scope of the project

1. Objectives

- To study the input parameter required for implant stability measurement in terms of micro-motion/resonance frequency.
- Finite element analysis of dental implant prosthetics.
- To develop a relation between micro-motion/resonance frequency and implant stability quotient at the implant-bone interface.
- Verification of the results with existing literature.

2. Problem Statement

In Dentistry, determining primary stability plays very important role to avoid failure of dental implant. By determining stability one can reduce one's visits to dentist in terms reducing its overall cost required for implantation process. By determining primary stability of dental implant, dentists are provided with some useful information (Bone quantity/quality, early implant loading, etc.). Hence, determination of the implant stability quotient of the implant is very important in such a way that it helps the dentist to determine the micro-motions and any flaws present after the implantation.

3. Scope of the project

1. Proposed research work is based on in-vitro study only.
2. Proposed research work does not include any clinical trials, animal study (in-vivo study)

Chapter-4

Materials And Methods

4.1 MATLAB Simulink Simulation

4.1.1 Introduction

While developing any experimentation, it is mandatory to understand the working of the system and how the system will react under certain circumstances beforehand. Also, based on the results achieved in the simulation, one can decide the components and materials needed for the actual experimentation more precisely. In order to understand how the dental implant behaves under certain circumstances and boundary conditions, a MATLAB Simulink model mimicking the actual experimentation process was developed. The primary aim of developing a Simulink model was to understand the resonance frequency of the dental implant achieved when subjected to the varying input frequency of fixed amplitude and also when the characteristics of the cancellous and cortical bone surrounding the dental implant are taken into consideration. There were very few studies where the mathematical model of the dental implant was implemented. However, based on the physical experimentation carried out in the numerous studies [18, 19], we developed a Simulink version of the proposed experimental setup.

4.1.2 Spring mass damper system

In order to carry out the mathematical modelling of the stability measurement of dental implant system, the best approach is to convert the biological interface of dental implant and surrounding cancellous and cortical bone to a mechanical spring mass damper system. Since an external force is acting to measure the stability of the dental implant and the cancellous and cortical bone around provide considerable amount of damping, a single degree of freedom forced damped vibrations model is the best approach to represent the system under consideration. In a forced damped system, whenever a force F acts on the spring mass damper system having mass ‘m’, damping ‘b’ and stiffness ‘k’, displacement ‘y’ takes place. The equation 1 represents the equation for the system. The image of the spring mass damper system is shown in Figure 1.

$$m\ddot{y}+b\dot{y}+ky = F \qquad (1)$$

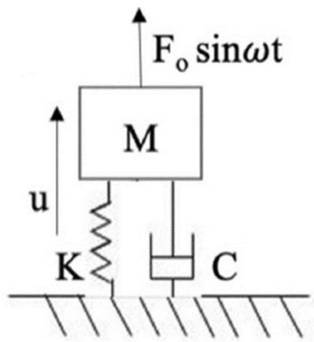


Figure. 10 Spring Mass Damper System [3]

4.1.3 Simulink Model

We used the software MATLAB Simulink in order to develop a virtual experimental setup. The model was a single degree of freedom spring mass damper system in which the mass mimicked the dental implant with abutment and fixing screw and the spring stiffness and damping mimicked the properties of the surrounding cancellous and cortical bone [3]. Referring to other study conducted in which function generator was used to generate a varying frequency, we used the virtual function generator block in Simulink to excite the mass with varying frequency [20]. The excitation frequency was varied from 2 kHz to 10 kHz from the function generator block. The resonance was measured with the help of the oscilloscope. The frequencies at which the maximum amplitude occurred was termed as the resonance frequency. The resonance Frequency is largely dependent upon the stiffness at the bone implant interface [21]. The parameters considered for the Simulink model are mentioned in Table 1. The complete image of the Simulink model developed can be seen in Figure 2.

Table 1: Parameters for Simulink Model

Mass	0.478*10^(-3) Kg
Stiffness [6]	7604 N/mm
Damping	30.144
Time	1 Second
Signal Variation	2 kHz - 10 kHz

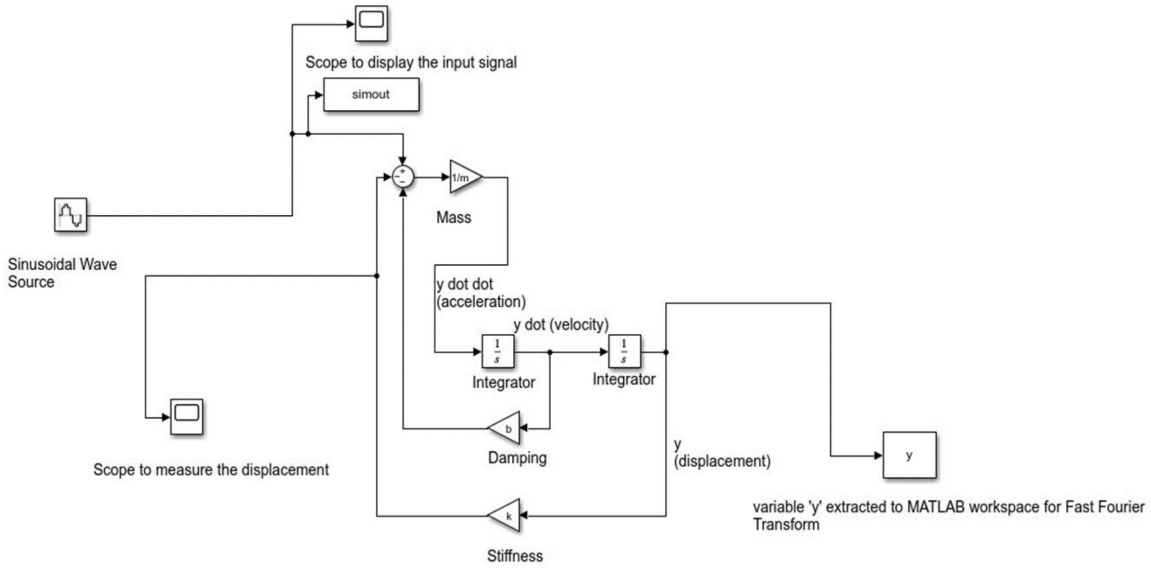


Figure 11: Simulink Model

Summary

The variables m (mass), k (stiffness) and b (damping) were assigned values using a MATLAB code. The resultant displacement was taken using the variable ‘y’. The results initially obtained from the ‘y’ variable were in the time domain. The amplitude of the time domain result is nothing but the displacement of the system in meters. Although the maximum displacement at the bone implant interface can be easily detected through the time domain signal, the frequency at which the maximum displacement is occurring is not understood. For understanding the frequency, the result has to be converted from time domain to frequency domain. The result in the time domain is shown in Figure 3 and the result in the frequency domain is shown in Figure 4.

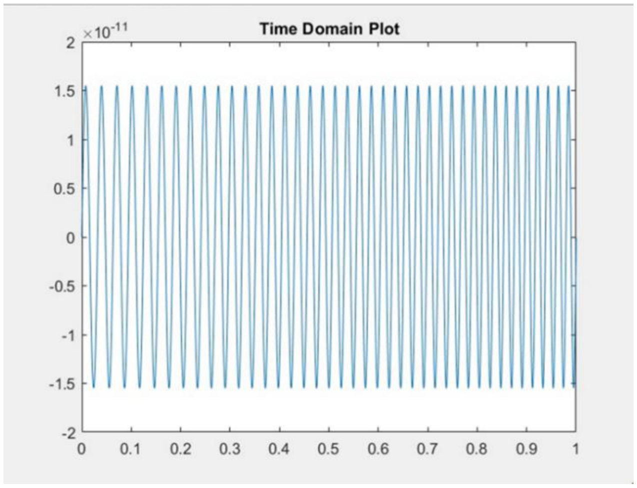


Figure 12: Results in Time Domain

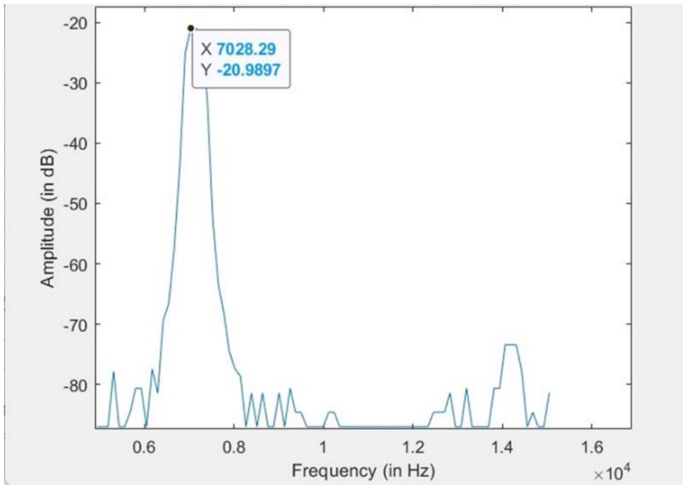


Figure 13: Results in Frequency Domain

From the plots of the results, it is evident that the maximum micromotion occurring at the bone implant interface is 1.5 * 10^{-11} m and the frequency at which the maximum micromotion is achieved is 7028.29 Hz. The amplitude achieved in the frequency domain is -20.9897 dB. From the studies conducted in the past, the resonance frequencies of the implants which are stable ranged from 7000 Hz to 10000 Hz [22,23]. So, it can be safely asserted that the conditions under which the simulation was carried out was the situation in which the dental implant was completely stable.

4.2 CAD Modelling

The cad modelling for this project is performed in Fusion 360 Software. The following components of the dental implant system were modelled accordingly.

4.2.1 Dental Implant:

The dimensions for the dental implant are derived from the Optical profile measurement machine. Images of these bio-horizon implant are clicked using the machine and further they are traced in the CAD software. The implant chosen for this project is a one start implant screw with a pocket for abutment and intermediate screw fixture. This implant has two cutting flutes which is a necessity for proper osteointegration. The exact values for the dimensions from these tracings are considered and the dental implant are modelled. The dimensions for the implant design are noted in the table below,

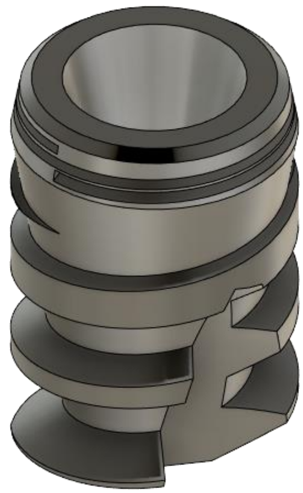


Figure. 14**:** Dental implant

Table No. 2**:** Dental implant dimensions

Parameters	**Dimension**
Top surface diameter	3.66 mm
Bottom surface diameter	2.65 mm
Thread Pitch length	1.60 mm
Height of the implant	6.27 mm

4.2.2. Abutment and intermediate screw

The abutment that has a perfect fitment over the implant for placement of crown or smart peg for measurement of stability is designed with dimensions that allow exact contact with the implant internally from the top surface of the implant. The implant and abutment are internally locked with a hexagonal locking joint and further both are rigidly fastened by an intermediate screw. The abutment is a hollow component which is required for fitment of internal component such as the intermediate screw. The dimensions for the Abutment design and intermediate screw are noted in the table below,

Figure. 15: Abutment

Table No. 3: Abutment dimensions

Sr. No.	Parameters	Dimension
1	Hole Diameter for intermediate screw	0.8 mm
2	Hexagonal side length	0.638 mm
3	Taper angle for connecting pipe	14°
4	Diameter for resting part on the implant top surface	5.20 mm
5	Taper angle for parent body of abutment	3.5°
6	Diameter for top surface of the abutment	3.60 mm

4.2.3 Crown

The design for crown that will be placed on the abutment is generated using foam tool in the CAD software. With this foam tool one can model a desired shape with any bends and curves. For this current research project, the tooth selected is molar type and this is carried out to satisfy the abutment design. The design of such molar crown is dependent on the external diameter of the abutment.[24]

4.2.4 Bone block

As mentioned in the above table the bone block comprises of two materials cancellous and cortical. A rectangular block with dimensions for both cancellous and cortical is mentioned in the table below. The different bodies in CAD design for cortical and cancellous bones are a necessity so as to carry out the FEM analysis using varying densities. [25]

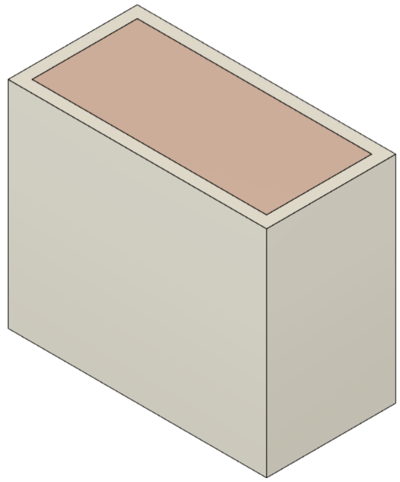


Figure. 16: Bone block

Cancellous

Table No. 4: Cancellous Bone block dimensions

Parameters	**Dimension**
Length	28 mm
Width	13 mm
Height	23 mm

Cortical

Table No. 5: Cortical Bone block dimensions

Parameters	**Dimension**
Length	30 mm
Width	15 mm
Height	25 mm

4.2.5 L-Shaped transducer:

As stated above the L Shaped Transducer consists of 2 Piezoelectric Elements and the L shaped metal strip. The diameter for the piezoelectric elements was identified as 27 mm, and for the L shaped metal strip the dimensions were considered as 10 mm width, 30 mm and 45 mm length for the short and long arms respectively.[26]

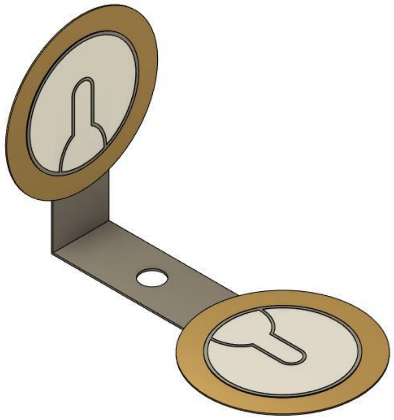


Figure. 17: L-shaped transducer

4.2.6 Assembly

All the components are imported in a single design and translated to their respective positions. The bodies in the assembly are converted to components in Fusion 360 for application of materials to the component. This assembly will be further imported in Hypermesh Software to carry out the modal analysis.

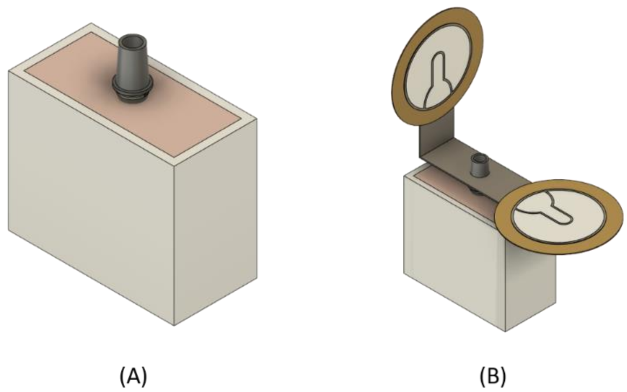


Figure. 18: The above figures depict (A) Bone bock assembly without L-Shaped Transducer (B) Bone Block with L-Shaped Transducer

4.3 Finite Element Analysis:

Finite Element method is a numerical method used for the solution of wide range of engineering problems. It is being extensively used in analysis and design of many complex real-life systems. Finite Analysis method finds application in deformation and stress analysis, vibration analysis and heat flow analysis. The purpose of FEA is

- To find displacement at particular points
- To find entire distribution or displacement field
- Calculate stress distribution
- Calculate natural frequencies and related mode shapes
- Find temperature distribution and hence thermal stress distribution

FEA is computational technique used to obtain approximate solution of boundary value problems in engineering. The boundary value problem is a mathematical problem wherein one or more dependent variable must satisfy a differential equation everywhere within a known domain of independent variables and thus it has to satisfy condition of the boundary of the domain

4.3.1 Finite Element Terminologies:

- Element: The approximate solution of boundary value examples in engineering are further converted into smaller regions called elements
- Nodes: The elements are further interconnected at specified regions called nodes
- Degree of freedom: The number of independent co-ordinates must be specified to define all displacement.
- Loads and constraints: Loads are applied forces that will occur during the use of component. It is used to know the failure modes of the component.

Applications of Finite element analysis

The applications of finite element analysis are as follows

- Crack and fracture analysis of dynamic loads
- Automotive applications
- Manufacturing process simulations
- Aerospace applications
- Stress analysis of truss and frame
- Biomechanics

Advantages

- Various types of load handling can be done
- Can be implied for any irregular shapes
- FEM can be used for coupled analysis

Disadvantages

- The cost required to acquire the solution is high
- The stress value may vary due to the quality of mesh
- Experts' assistance is required for the interpretation of the results
- Aspect ratio may affect the final results.

The Finite Element (FE) model (implant block assembly) taken under the consideration for in-vitro study represents the pre-molar teeth placed into the mandible. The model is tetra meshed with (number of nodes) in Hypermesh software. [27] The computer aided 3D model is further analyzed by applying the required boundary conditions for modal and dynamic analysis to calculate the results of the resonant frequency using Optistruct. Optistruct is user friendly and specifically designed to solve large and complex problems. Optistruct uses basic types of analysis such as eigen value extraction, frequency response analysis and transient response analysis by which any complex problems can be solved.

4.3.2 Material properties

finite element method (FEM) software is commonly used for the biomedical applications for the effective and successful design and analysis of dental implants and their responses. The representation of the cortical bone for the 2.0mm bone block acrylic material of density 0.3 is used while polyurethane material block of density 0.3gm/cm3 is used to mimic the porous cancellous bone present inside the cortical bone block. The titanium implant from the manufacturer Biohorizon is taken for experimentation and observing the resonance frequency of the implant when subjected to sinusoidal wave. The entire system is virtually created using CAD software and according the material properties are given to the system as shown in the table

Component	Material	Density(gm/cm^3)
Cortical	Acylonitrile butadiene styrene	1
Cancellous	Polyurethane	0.3
Implant	Titanium	4.5
Abutment	Titanium	4.5
Crown	porcelain	2.3

Table 6. Components along with material properties and their respective density

4.3.3 Meshing

The dental implant prosthetic is typically meshed with quadratic tetrahedral mesh with (no of elements) with the mesh size ranging between 0.07mm to 0.1mm. This bone model consists of two distinct components namely cancellous and cortical in which the cancellous part is meshed finely in order to maintain the accuracy for the results that are attained at the implant thread surface and the bone the cortical region away from the implant thread interface is coarsely meshed to sustain the size and shape of the bone block model. Figure shows the meshed implant assembly as well as the meshed components.

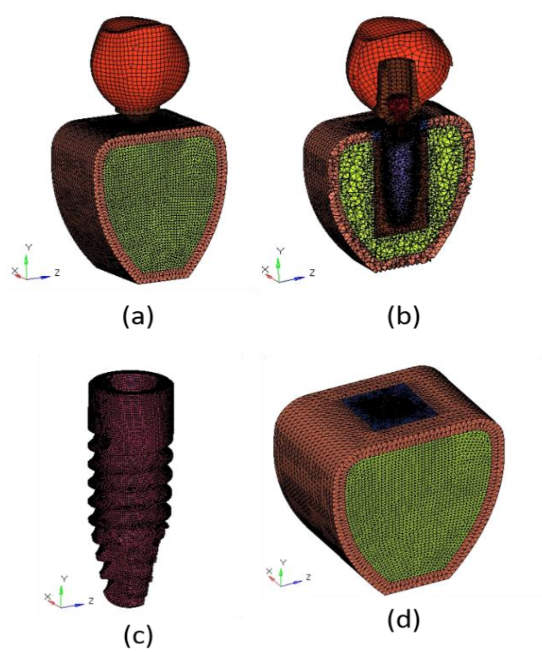


Figure 19: (a) shows the meshed implant assembly, 10(b) section view of meshed assemble, 10(c) meshed implant, 10(d) Meshed bone block

Bone Block

This model comprises of two distinct components namely cancellous and cortical. Out of which the cancellous part is meshed finely with regards to the accuracy for the results that are attained at the implant thread interface and the bone. The element size for the bone block was selected on the basis of the convergence criteria. For the analysis mesh size of 0.1mm was considered. Further the cortical region away from the DIP thread interface is coarsely meshed to sustain the size and shape of the bone block model for each material. The number of elements can be referred from Table 1

Crown

The crown is modelled using linear tri/quadrilateral elements of 0.3 mm in size. The meshing is done as a 2D mesh as no critical points are present for the measurement in the current analysis on the crown interface. The number of elements can be referred from Table 7.

Table 7: Number of elements and nodes present in individual components

	Cancellous Bone	**Cortical Bone**	**Implant Model**	**Abutment**	**crown**	**Total**
Number of Nodes	68475	22904	44210	4973	1318	137080
Number of Elements	344872	103760	206545	19963	1340	687555

4.3.4 Boundary condition/ Modal Analysis

Modal analysis is performed on the bone block assembly wherein the lower central node is fixed and the rest of the assembly is allowed to vibrate at its natural frequency. The natural frequency is directly proportional to the stiffness of the material. Greater the natural frequency greater will be the stiffness of the material. [28] Modal analysis is used to monitor and measure the vibration of each component at its natural frequency. Modal analysis can clarify the vibration of each part of the structure in a resonant state by vibrating an actual structure using an exciter or an impulse hammer. Using a modal analysis software to fabricate the wire-frame of the structure, we can investigate how the structure vibrates at the natural frequency. [29,30] The natural frequency of vibration is the frequency of a body due to its own elastic properties. Hence this natural frequency is important for the resonance frequency analysis to determine the stability of the dental implant or the degree of osseointegration. Figure shows the boundary condition applied on the implant for the modal analysis to find out modal frequencies.

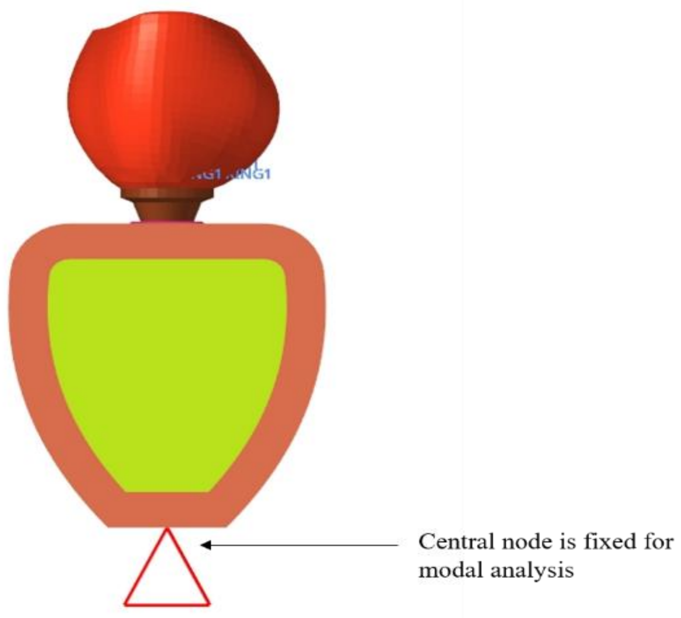


Figure 20. Implant assembly with applied boundary condition for modal analysis

4.3.5 Convergence criteria for elements

The convergence requirements are met by refining components at the contact surface in implant systems. These parameters are taken into consideration in order to ensure that the component of the implant system that is essential for the execution of the analysis has a greater number of parts, which will lead to more precise findings. The variance in the findings should not be higher than 5% for a successful mesh convergence

4.4 Experimentation

The basic principle behind the complete experimentation is that the dental implant is vibrated within a certain frequency range and the specific frequency at which the dental implant vibrates with maximum amplitude is termed as the resonance frequency. For vibrating the dental implant with certain frequency, a sinusoidal waveform is required. To fulfill this purpose, a function generator is used. Function generator is a device which is used to generate a wide range of electrical waveforms [31]. In our experimentation, the frequency range is between 2 kHz to 10 kHz. Generally, the local maxima within a specific frequency range at which the maximum amplitude of vibrational response is generated is considered as the resonance frequency of the dental implant during that trial. The dental implant, along with the L-shaped piezoelectric transducer, was immersed in the dental plaster mixture, which at the initial stage is in a form of a paste. The initial condition mimics the exact condition of an unstable dental implant.

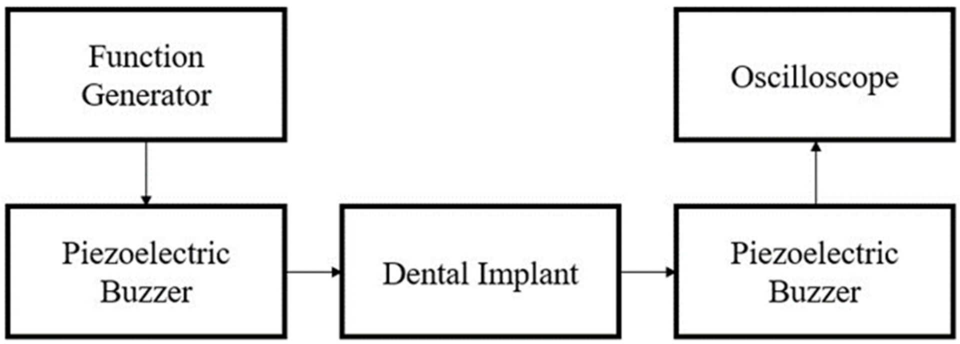


Figure 21: Experimentation Flowchart

The initial reading of frequency measurement is taken immediately after the implant is placed in the dental plaster mixture. Over the period of time, the dental plaster hardens and this process simulates the osseointegration of the dental implant taking place at the bone implant interface. Overall, 10 readings are taken at fixed intervals right from placing the implant in the dental plaster mixture until the dental plaster hardened completely.

In the piezoelectric transducer, it shall be noted that the first piezoelectric buzzer is connected to the function generator and is responsible for exciting the dental implant whereas the second piezoelectric buzzer is responsible for measuring the vibrational response of the dental implant and further displaying the response on the oscilloscope. At each instance when the frequency is measured, the response is displayed on the oscilloscope. Initially, the waveform and the peak-to-peak voltage which, in this case, is the amplitude of the waveform is displayed on the oscilloscope. The instance at which the maximum amplitude of the vibrational response occurs can be visualized on the oscilloscope on this mode, however, the frequency at which this maximum amplitude occurs cannot be visualized. Hence, in order to understand the frequency, the Fast Fourier Transform (FFT) function inside the oscilloscope is used. The complete flowchart of the experimentation can be seen in Figure 5. The complete experimental setup can be seen in Figure 6.

Figure 22: Experimental Setup

4.5 Measurement of Implant Stability Quotient (ISQ)

In the year 1996, the concept of RFA was developed by Meredith et al [32] who also developed the first generation setup of the device measuring the stability of dental prosthetic. However, the first-generation setup consisted of a lot of wiring and was very convoluted to use. Hence, in the year 1999, Ostell device was designed by Integration Diagnostics Ltd. There have been total seven generations of this developed until now. Although all these devices work on the RFA principle, Ostell had developed a new scale called Implant Stability Quotient (ISQ) which ranged from 0 to 100, 0 being the least stable value and 100 being the most stable value. All the Ostell devices run on the ISQ scale. The device required a transduce to be attached on the healing abutment of the dental implant which is excited using electromagnetic force [33].

Recently, years after Ostell devices were developed, a new device, called Penguin RFA was developed by the same researchers who developed the Ostell devices. The major novelty introduced in this device was that a new transducer, called Multipeg, was used which was claimed to be easier to use and most importantly this transducer was reusable [11].

Since the goal of the project was to establish the relationship between ISQ and RF, measuring RF and ISQ simultaneously was necessary. The Penguin RFA device was used to measure the ISQ of the dental implant at fixed intervals.

The following components were used is this project

- Signal generator
- Piezoelectric transducer
- L-shaped metal strip
- Artificial Bone blocks
- Oscilloscope

1. Signal generator-

An electronic device or instrument that generates continuous and discrete signals like analog and digital is known as a signal generator. A signal generator is used to generate a specific frequency. There are various types of waveforms that can be generated using signal generator, these include triangular, square, ramp, pulse and sine. It can generate frequencies in wide bandwidth. The signal generator block diagram is shown below. In the block diagram, voltage-controlled oscillator is an essential part because the input-controlled voltage can be determined through the frequency of the voltage-controlled oscillator. So both the control voltage and VCO's frequency are directly proportional [33].

Once the signal is given to the control input then it generates the oscillator frequency. Once the audio input signal is given to the control voltage then the frequency-modulated signal will be produced using VCO. The signal generator generates the tone, waveforms in digital patterns, and arbitrary. Once the signal generates an unmodulated signal, then these are known for producing continuous wave signals. So, it generates a square wave modulated signal, complex and triangular waves, etc.

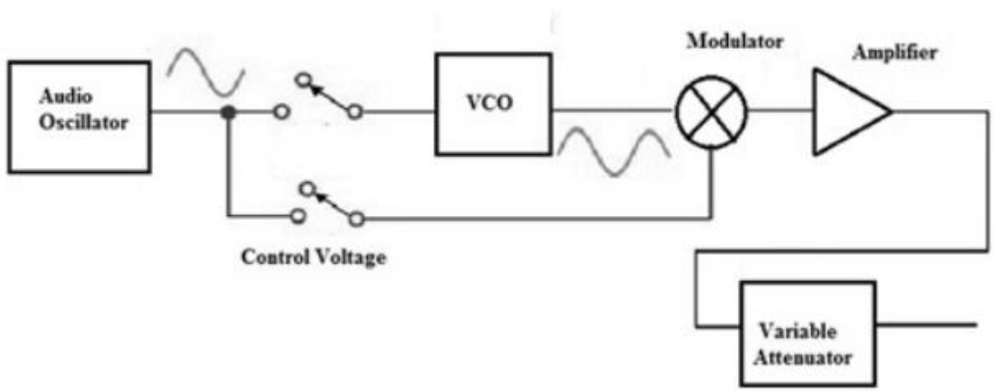


Figure. 23. Block diagram of signal generator circuit [33]

In this project signal generator is used to generate sine wave in the range of 3kHz to 10kHz 3 and peak to peak amplitude of 1 volt. This waveform is sent to one of the piezoelectric transducers. The input frequency is modulated carefully in stepwise manner.

Table 8. Signal Generator Specifications

Sr. no	Description	Caddo 4061
1	Operating modes	Sine, Square, Triangle, Ramp, Pulse, TTL
2	Frequency range	0.2 Hz-2 MHz in steps
3	Frequency accuracy	± 0.5%
4	Output Impedance	50Ω
5	Output voltage	20 V O.C
6	Resolution	1Hz
7	Sensitivity	0.5V
8	Input Impedance	1M Ω
9	Operating temperature	0 to 40°C

Figure 24. Signal Generator

2. Piezoelectric buzzer-

A piezoelectric buzzer is a disc shaped transducer, which works on the principle to piezoelectric effect. The piezoelectric effect states that Piezoelectric Effect is the ability of certain materials to generate an electric charge in response to applied mechanical stress. The word Piezoelectric is derived from the Greek piezein, which means to squeeze or press, and piezo, which is Greek for "push".[34]

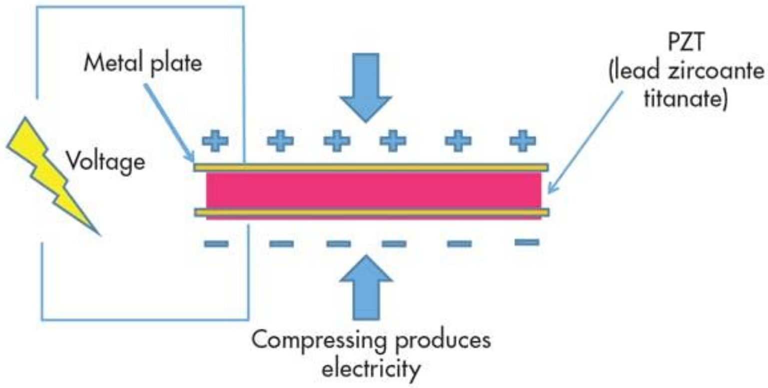


Figure 25. Piezoelectric effect [34]

One of the unique characteristics of the piezoelectric effect is that it is reversible, meaning that materials exhibiting the direct piezoelectric effect (the generation of electricity when stress is applied) also exhibit the converse piezoelectric effect (the generation of stress when an electric field is applied). When piezoelectric material is placed under mechanical stress, a shifting of the positive and negative charge centers in the material takes place, which then results in an external electrical field. When reversed, an outer electrical field either stretches or compresses the piezoelectric material. [35]

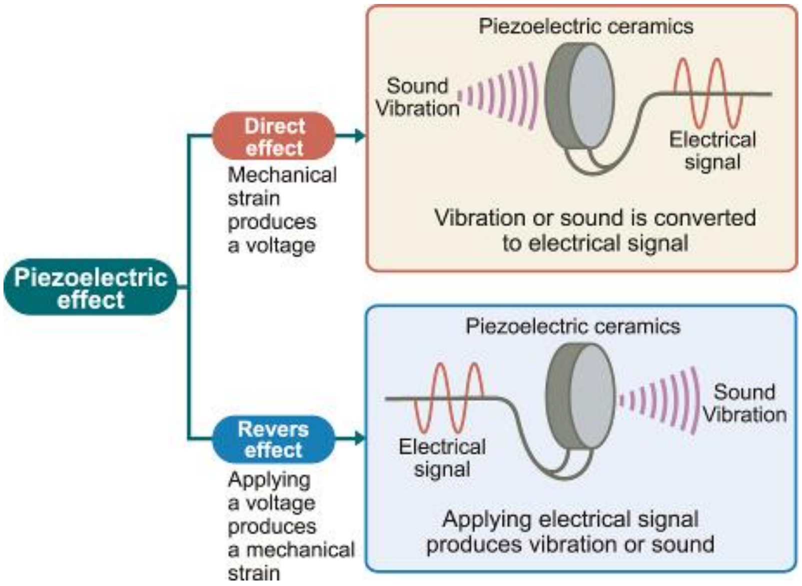


Figure 26. Piezoelectric transducer [35]

There are many materials, both natural and man-made, that exhibit a range of piezoelectric effects. Some naturally piezoelectric occurring materials include Berlinite (structurally identical to quartz), cane sugar, quartz, Rochelle salt, topaz, tourmaline, and bone (dry bone exhibits some piezoelectric properties due to the apatite crystals, and the piezoelectric effect is generally thought to act as a biological force sensor). An example of man-made piezoelectric materials includes barium titanate and lead zirconate titanate.[36]

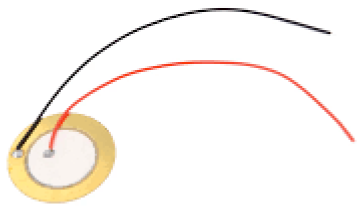


Figure 27. Piezoelectric buzzer

In this experiment, a piezoelectric buzzer was used to generate specific vibrations according to the input sine wave. The implant is vibrated with certain frequency. When the input frequency matches natural frequency of the implant resonance occurs. The second piezoelectric transducer captures this resonance frequency. The output of this piezoelectric transducer is sent to the oscilloscope.

The two piezoelectric discs are mounted on an L-shaped metal strip made up of tin. The L-shaped metal strip was manufactured in college workshop with dimensions of the piezoelectric buzzers taking into consideration. [37]

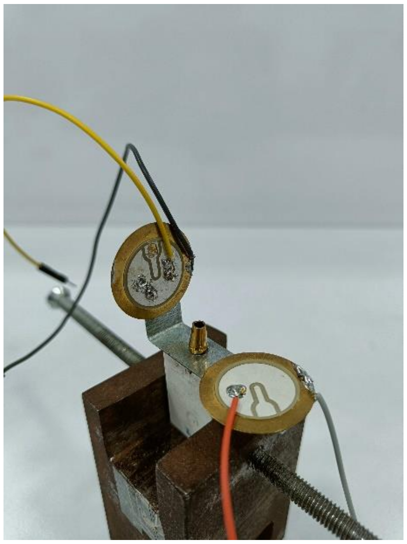


Figure 28. Piezoelectric buzzer mounted on L-shaped strip

3. Dental Implant system and Bone block-

Dental implants have been popular for option for teeth replacement. Dental implants come in various sizes and designs, they can be short and long and with varying diameter. Suitable implant is selected according to the patient's oral health. In this experiment implant of biohorizons make is used. An abutment is fixed on the implant with the help of a screw. It acts as a bridge between dental implant and crown. Abutment is generally made up of Titanium alloy.

Implant properties-

Table 9. Implant Specifications

Sr. no.	Parameter	Value
1	Material	Titanium alloy (grade 5)
2	length	15 mm
3	Diameter	3.5 mm
4	Density	4.43 g/cc

Bone blocks-

As the study is in-vitro, there is a need of creating an artificial bone model that will mimic the properties of human jawbone. The bone in this region is made up of two different regions, which are cortical and cancellous. Cortical being the outer and harder bone and cancellous being inner and soft bone. To replicate cortical bone acrylonitrile butadiene styrene (ABS) is used and to replicate cancellous bone polyurethane blocks are used. The hard outer layer of the bone block was manufactured using 3D printing technique. To replicate the inner cancellous bone dental plaster is used. To prepare the dental plaster mixture, 8 grams of plaster mix is added to 4 cc of water. From this, a paste like consistency is achieved. As time passes, the dental plaster sets and solidifies this process mimic the osseointegration phenomenon. [38]

Table 10. Specifications of cortical bone (ABS)

Sr. No	Parameter	Value
1	Thickness	2.0 mm
2	Density	1.05 g/cm3

Table 11. Specifications of cancellous bone (dental plaster)

Sr. No	Parameter	Value
1	Density	0.4 g/cm3

Figure 29. Bone Block

4. **Oscilloscope-**

An oscilloscope is a laboratory instrument commonly used to display and analyze the waveform of electronic signals. In effect, the device draws a graph of the instantaneous signal voltage as a function of time. The oldest form of oscilloscope, still used in some labs today, is known as the cathode-ray oscilloscope. It produces an image by causing a focused electron beam to travel, or sweep, in patterns across the face of a cathode ray tube (CRT). More modern oscilloscopes electronically replicate the action of the CRT using a liquid crystal display (liquid crystal display) similar to those found on notebook computers. In this experiment we used Tektronix TDS2001C oscilloscope to view the results. The specifications of the oscilloscope are given below.[39]

Table 12. Specifications of oscilloscope

Sr. No.	Parameter	Value
1	Bandwidth	50MHz
2	Channels	2 (CH1, CH2, Ext, Ext/5, AC Line)
3	Sample rate	500 MS/s
4	Waveform Math	Add, Subtract, Multiply, FFT
5	Input impedance	1 MΩ in parallel with 20 pF
6	Position range	2 mV to 200 mV/div +2 V; >200 mV to 5 V/div +50 V
7	Peak detect	High frequency and random glitch capture.
8	Reference waveform display	Two 2.5k point reference waveforms

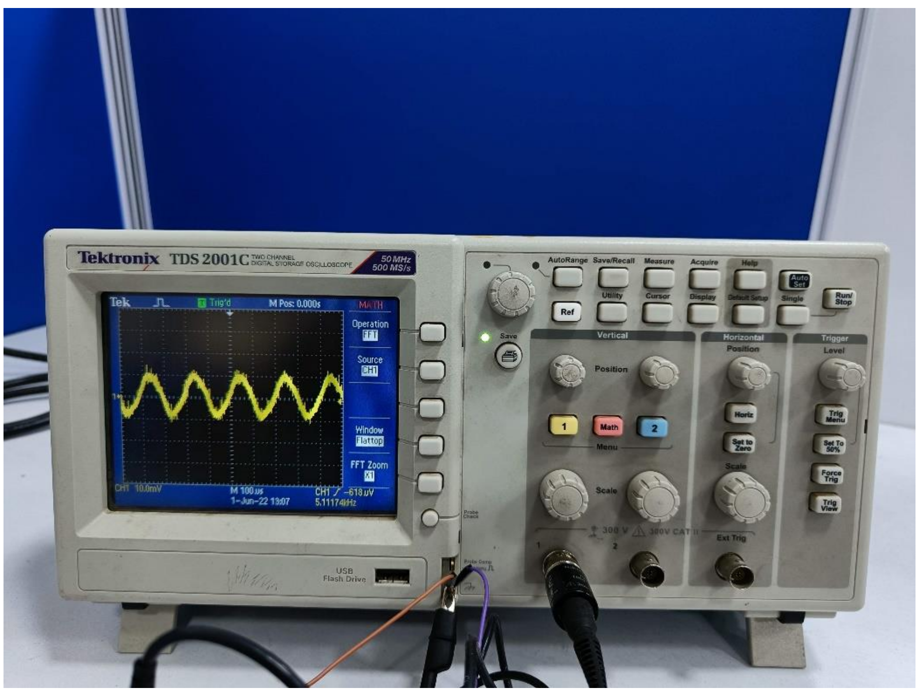


Figure 30. Oscilloscope [39]

4.5 Relationship between Implant Stability Quotient and Resonance Frequency

Implant Stability Quotient (ISQ) is the prevalent unit of measurement of dental implant stability in all the devices which work on the principle of RFA. The scale ranges from 0 to 100. The stability of the dental implant in linearly correlated with the ISQ scale. Generally, the ISQ value below 40 is termed as risky and there is a high probability that the implant will fail [40]. The value of ISQ is basically a tool for dental surgeons which helps them to decide when to load the dental implant. Although the minimum ISQ value for loading the dental implant is still under conflict, overall a value of 70 is considered as a fair value to initiate implant loading [13].

Tracking ISQ values throughout the healing process will often demonstrate a decrease in stability over the first three weeks following implant placement. Clinical research has demonstrated the lowest values and most significant reductions in stability occur at sites with poor bone quality. Any observed decrease in stability is thought to be related to bone healing following surgical injury. Understanding the healing process and its effect on implant stability can assist in clinical decision-making related to the timing of implant restoration. In this regard, RFA can provide an objective assessment of implant stability, bone healing and osseointegration [41].

Although the ISQ scale ranges from 1 to 100, implant ISQ values generally range between 50 and 80. When translating ISQ values into implant stability, it is important to note ISQ values share a nonlinear, inverse relationship with implant micromotion. For example, an increase of ISQ from 60 to 70 reflects a 50% reduction of implant micro mobility. Looking at certain benchmarks for ISQ and implant stability, the following guidelines are often employed: ISQ ≥ 70 = high stability; ISQ 60–69 = medium stability; ISQ < 60 = low stability (Figure 5). Rather than considering a single ISQ value, however, changes or trends in ISQ over time can provide valuable information regarding development of secondary implant stability or osseointegration. Successful implants tend to demonstrate increasing ISQ values during healing, particularly during the first six weeks following surgical placement. Implants with initially low (< 60) or medium (60–69) ISQ levels often demonstrate increases over time, while implants with initially high (≥ 70) ISQ values may remain stable or increase over time.

There is a strong evidence that suggests that the higher the RF of the dental implant, higher is the dental implant stability [40, 41]. However, the exact correlation between the ISQ and RFA value is still not known. In order to understand the exact correlation between ISQ and RFA and also to aid in further development of RFA based medical devices, we have derived the correlation between ISQ and RFA.

1.5.1Linear Regression

Linear regression is a statistical approach which models the linear relationship between one scalar variable and two or more explanatory variables. This approach uses linear predictor functions whose unknown parameters are found out using the existing data. The procedure of linear regression contains one dependent variable and one or more independent variables. The general formula for linear regression looks like follows:

$$y_i = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_{ip} + \varepsilon_i \text{ , } i = 1,\ldots,n \quad (2)$$

where y is the dependent variable, x is the independent variable, β is a (p+1) dimensional parameter vector and ε is called as the error term. The plot of linear regression contains data points on the plot and also a trendline, of which the slope and position is calculated using statistical techniques. A trendline basically indicates whether the values over the whole data set are increasing or decreasing in value. One major advantage of using the linear regression algorithm was that there is very less chance of overfitting, thereby improving the predictability of newer values in the future. In the experimentation conducted, overall 10 readings of RF and ISQ values were considered and further the linear regression was carried out on the dataset given in Table 4.

Table 13: Dataset used for linear regression

Resonance Frequency (Hz)	Implant Stability Quotient
2151	35
2512	37
4000	57
4597	60
5432	68
5600	70
6075	70
6201	72
6531	72
7114	83

Chapter-5

Results and Discussions

5.1 Experimental Result

The input frequency is varied in the range of 2 kHz to 8 kHz. Resonance frequency is an inarguable indicator of the stability of the implant. Higher the magnitude of resonance frequency, more the stability of the dental implant [42]. The experimentation conducted by Meredith et al [33] yielded the resonance frequency of a stable dental implant around 6900 Hz. In this experimentation, there was a decreasing trend of resonance frequency as the exposed length of the dental fixture increased. In the trials conducted by Ghosal et al [43], the results were measured using magnitude as well as phase plots of impedance. Initially, when the implant was just dipped in the dental plaster, the resonance frequency peaked suddenly and then as the dental plaster hardened, it stabilized at around 7.3 kHz. The trial conducted by Baruah et al [44] yielded the resonance frequency of a stable implant around 8500 Hz. The experimentation conducted in [45] consisted of experimental as well as numerical analysis of five different types of dental implants, in which, type 1 exhibited the highest stress levels and type 3 and 4 exhibited the lowest stress levels. The trials conducted by [46] revealed a linear correlation between trabecular bone density and resonance frequency of the Orthodontic Mini Implants (OMI) and also linear correlation between cortical bone thickness and resonance frequency of the OMI. The experimentation conducted in [47] aimed to provide the correlation of Implant Stability Quotient (ISQ) with the cortical bone thickness and radiographic bone density. It was reported in [18] that as the cortical bone thickness and radiographic bone density increased, the ISQ value increased as well, thereby suggesting that higher the ISQ, more stable is the dental prosthetic. [48] had stated that as the bone density increased, the resonance frequency also increased linearly.

During our experimentation, the maximum amplitude of the sinusoidal signal achieved was visualized on the oscilloscope. Also, in order to understand the frequency at which the maximum amplitude was achieved, FFT algorithm was carried out and the results were displayed on the oscilloscope.

Sanghamitra Ghosal et al [49] performed similar experiments with piezoelectric sensor on a single bone block. They varied the input frequency in the range of 6.5kHz to 9.5kHz and keeping amplitude constant. Their results showed that the resonance frequency in the range of 5kHz- 8.5kHz. Similarly in this experimental study the input frequency was varied from 3kHz to 10kHz while keeping the amplitude constant.

Sunil et al[50]. described similar tests in which the voltage outputs of similar cross-shaped transducers with 3-terminal piezo-elements were analysed to find the resonance frequencies. According to the findings of these investigations, the nominal resonance frequency of these systems is about 7 kHz. This finding was confirmed in the current situation, and hence the responses were obtained in the frequency range of 6.5 kHz to 9.5 kHz.

L. pagliani et al [51] performed experiments to find out RFA and micro-motion of dental implants. They used bovine bone samples and drilled into it according to the implant size. A total of 30 sites were prepared and implants were inserted with a specific insertion torque. RFA measurements were done by using osstell device using a smartpeg attached to abutment. Their results showed that when the ISQ values are high the micro motion in small and when ISQ decreases micro motion increases. Thus giving us the results for implant stability.

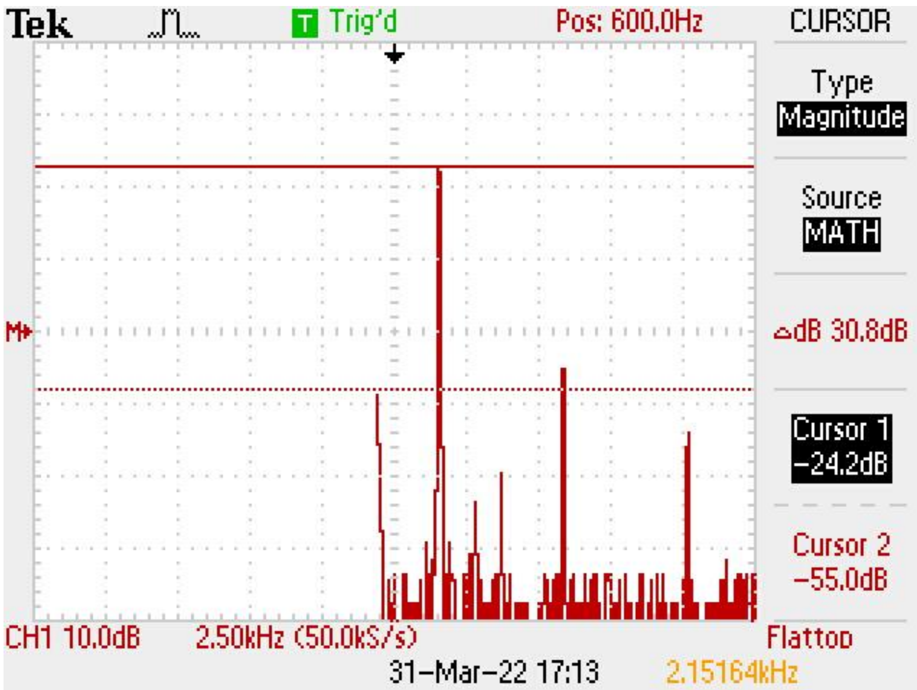


Figure 31 (a). graph for resonance frequency of 2.1516 kHz

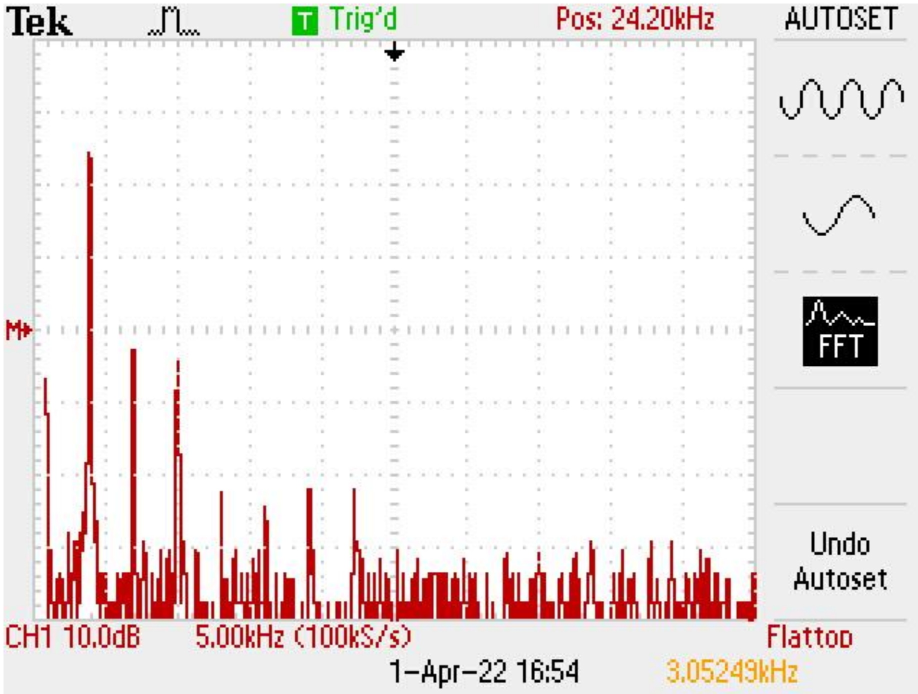


Figure 31 (b). graph for resonance frequency of 3.0524 kHz

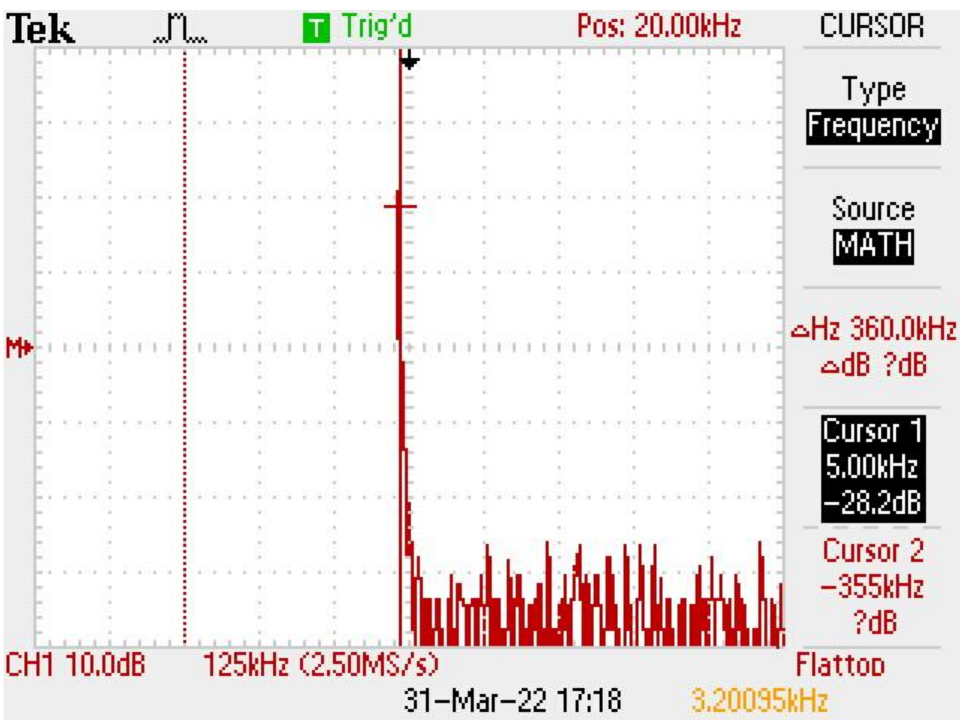


Figure 31 (c) Graph of resonance frequency analysys at 3.2009KHz.

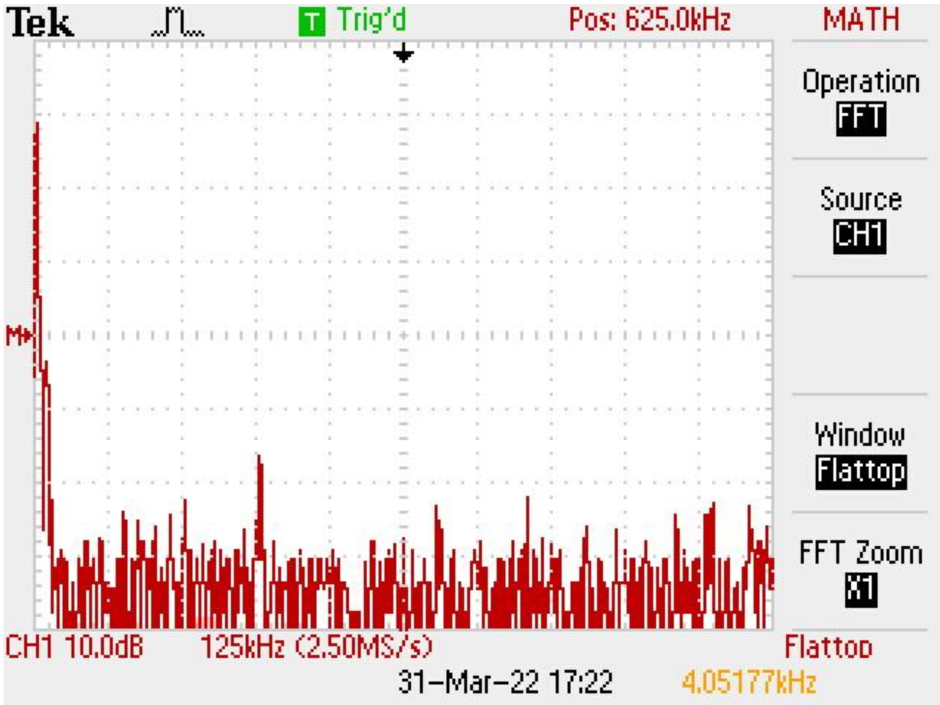


Figure 31 (d) Graph of resonance frequency analysis at 4.0517KHz.

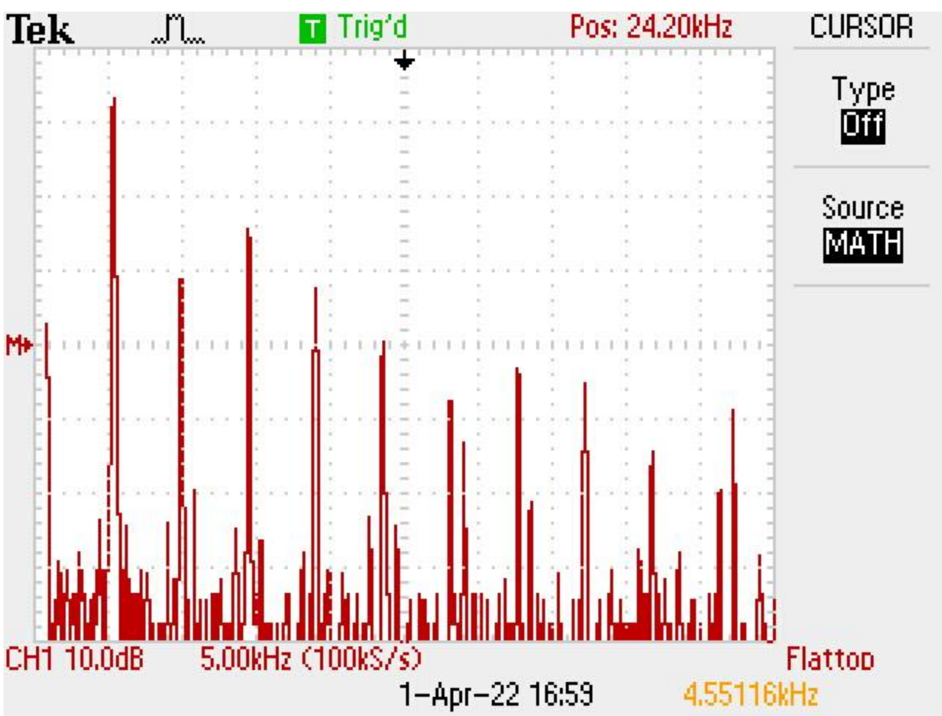


Figure 31 (e) Graph of resonance frequency analysys at 4.5517KHz.

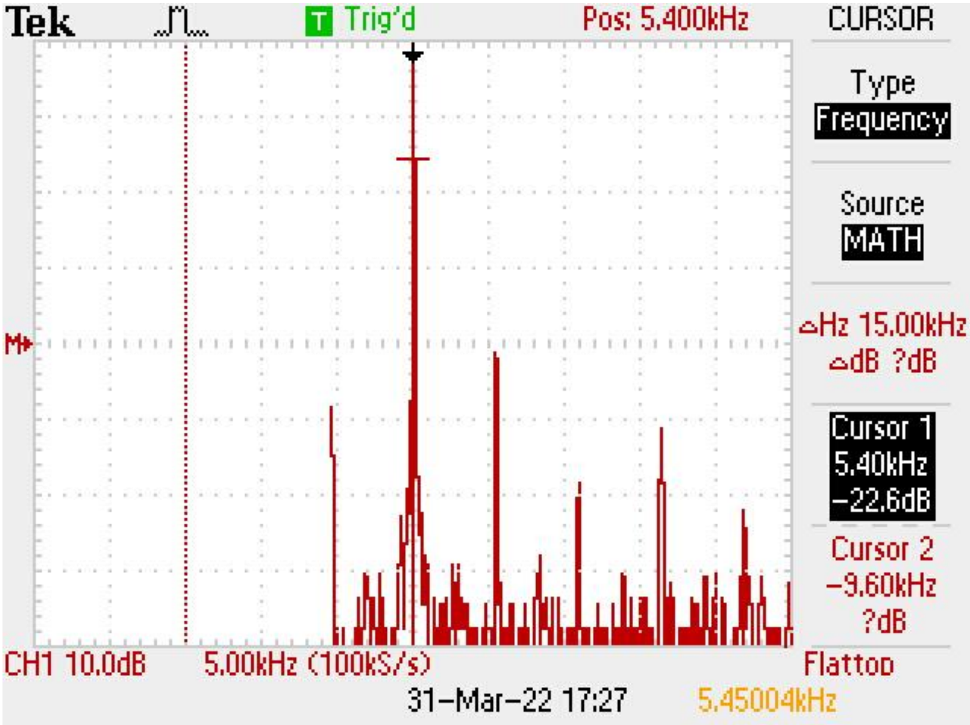


Figure 31 (f) Graph of resonance frequency analysis at 5.450KHz.

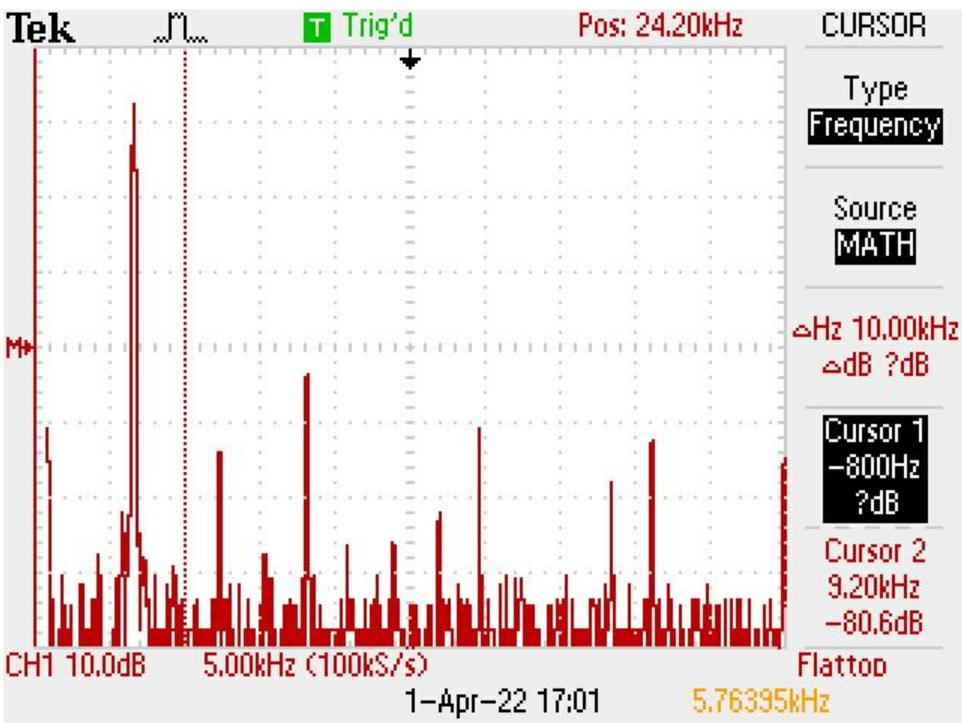


Figure 31 (g) Graph of resonance frequency analysis at 5.7639KHz.

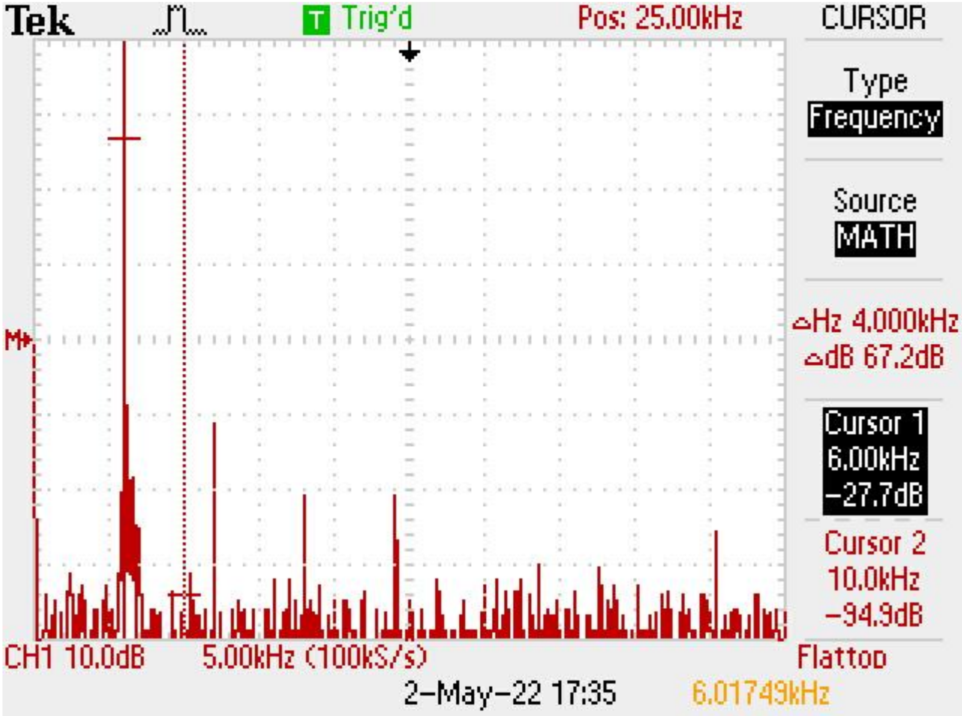


Figure 31 (h) Graph of resonance frequency analysis at 6017.4KHz.

The above FFT plots show that the resonance frequency lies in between the range of 2kHz to 7kHz. Further these results were used to find the correlation between resonance frequency and implant stability quotient.

There were 10 total readings taken at 2-minute intervals, making the complete trial of 18 minutes. The RF readings using the experimental setup and the ISQ readings using Penguin RFA were taken simultaneously. The different RF values with respect to time can be seen in Table 2 and the ISQ values with respect to time can be seen in Table 3. The plots for the same can be seen in Figure 7 and 8 respectively.

Table 14: Time vs Resonance Frequency (RF)

Time (Mins)	Resonance Frequency (Hz)
0	2151
2	2512
4	4000
6	4517
8	5432
10	5600
12	6075
14	6201
16	6531
18	7114

Table 15: Time vs Implant Stability Quotient (ISQ)

Time (Mins)	Implant Stability Quotient (ISQ)
0	35
2	37
4	57
6	60
8	68
10	70
12	70
14	72
16	72
18	83

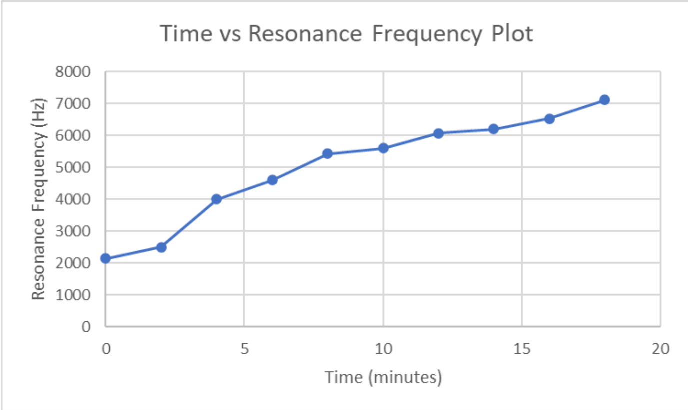


Figure 32: Time vs Resonance Frequency (RF) plot

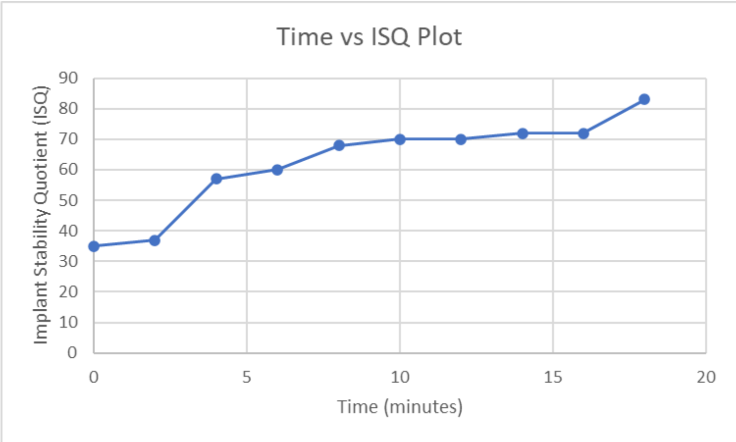


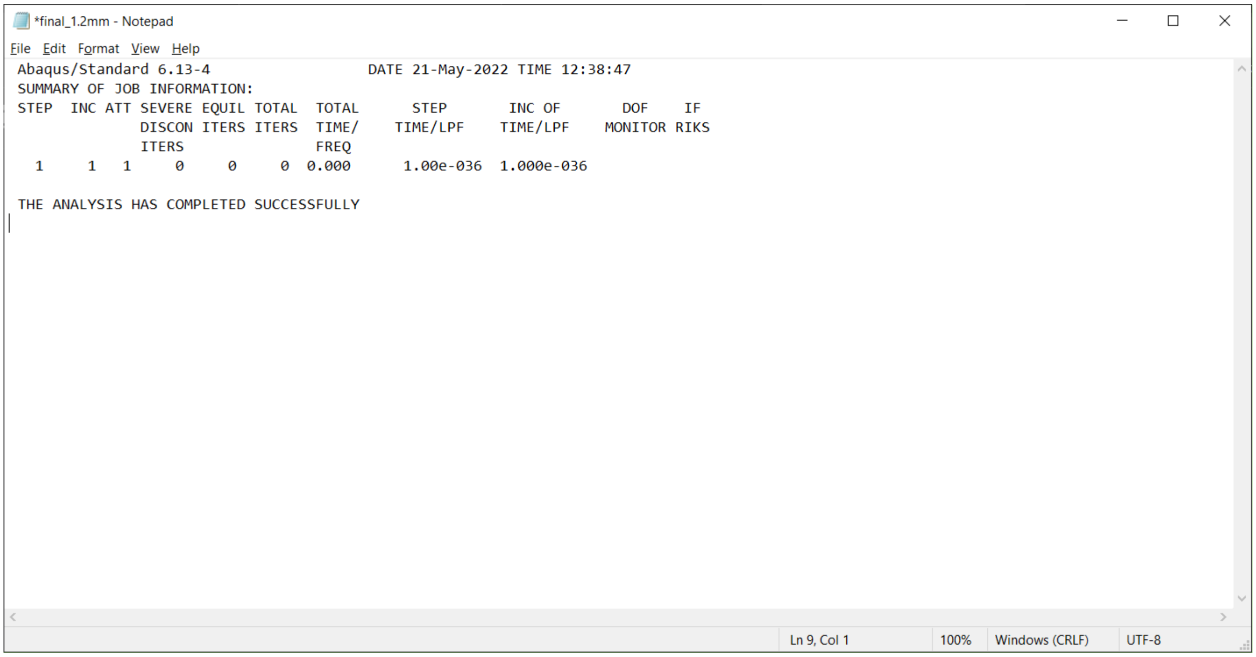
Figure 33: Time vs Implant Stability Quotient (ISQ) plot

5.2 Modal Analysis Results

For lateral and oblique loading situations, Dhatrak et al. Found the critical intensity of stress in the proximal area of the interface, but for vertical occlusal loading, it was found in the distal region. In comparison to vertical (occlusal load) and lateral loading circumstances, oblique loading (simulated as genuine masticatory loads) causes more stress at the interface. The stress intensity in a cancellous bone along the bone-implant contact is predicted using an experimental photoelastic methodology (Optical technique). [52] The research by Cairns et al found that when analyzing fixation, higher resonant frequencies and mode forms (rather than just the fundamental frequency) are useful. The model boundary conditions influenced the modal parameters. The estimated mode forms of 4 Nm (tight fit) and 0.5 Nm (loose fit) models for both boundary conditions were determined. The frequency difference between the 4 Nm and 0.5 Nm models is calculated in percentage. The fundamental frequency was found to be 10-47 percent lower for the 0.5 Nm insertion torque than the 4Nm insertion torque. [53] T. Sasaki et al [54] reported that the mean natural frequency in the splinted crown approximately 758 Hz was substantially higher than that in the non-splinted crown having 752.8 Hz natural frequency. Meanwhile the mean value of the greatest displacement was much less in the splinted crown (6.7 μ m) than that in the non-splinted crown (7.3 μ m). The vibration characteristics of the superstructures vary between designs with splinted and non-splinted crowns. Crown splinting enhanced the stiffness and natural frequency and lowered the MDP. Thus, our data imply that crown splinting lowers the deformation of the superstructure, implants, and the surrounding tissues in contrast to the deformation seen when no splinting is doneThe natural frequency and damping ratios were computed from the transfer function using modal analysis software, the modal shapes at each natural frequency were measured, and the maximal displacement with the lateral force during mastication was simulated.[55,56] Similarly the results that were obtain in this research gives us the maximum displacement of the dental implant system in lateral direction in y-direction and those are further stated below.

A. Suzuki et al performed modal analysis and claimed that implant prostheses with 6 abutments vibrated less than those with 4 abutments. The prostheses were excited using an impact hammer, and vibration was noted on each crown. Each subject's modal parameters were determined. Additionally, the modal forms at each natural frequency were compared. [57]

For this research after the meshing of the Dental implant prosthetic, and applying the respective boundary conditions as explained above modal analysis is carried out in Hyperworks using abaqus as the solver. The status file generated states that the Analysis run is completed successfully without any errors.



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SUMMARY OF JOB INFORMATION:
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               ITERS               FREQ
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THE ANALYSIS HAS COMPLETED SUCCESSFULLY
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Figure 34: Analysis Status file

The modal analysis done for this research project depicts the results given below:

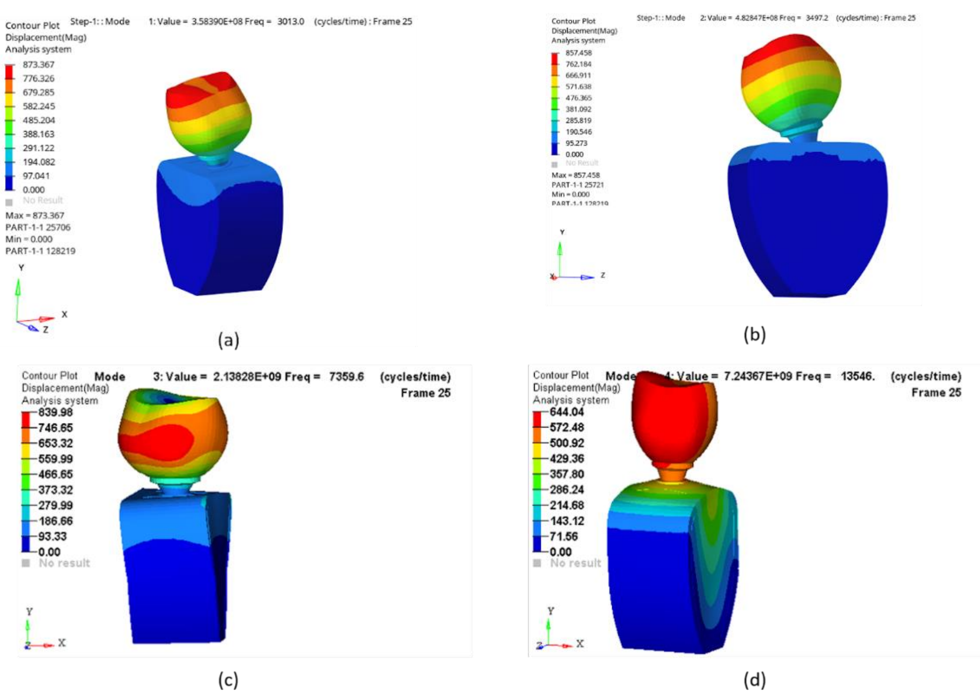

Figure. 35 (a) First mode with bending in x- direction, (b) Second mode with bending in z-direction, (c) Third mode with twisting in x-z plane, (d) Fourth mode with lateral displacement in y-direction.

The obtained modal frequencies are the dental implant system's inherent frequencies for the respective modes. The table below describes more on the above obtained result.

Table 16. Mode shapes and their respective frequencies

Mode number	**Mode type**	**Modal Frequency**
1	Bending	3013.0
2	Bending	3497.2
3	Twisting	7359.6
4	Lateral Displacement	13546

5.3 Co-relation Between ISQ and RFA

As the RF values increased with the increase in the hardness of the dental plaster, the ISQ values increased as well. When linear regression analysis was carried out, there was an increasing trend between both the parameters. The plot between RF and ISQ can be seen in Figure 9.

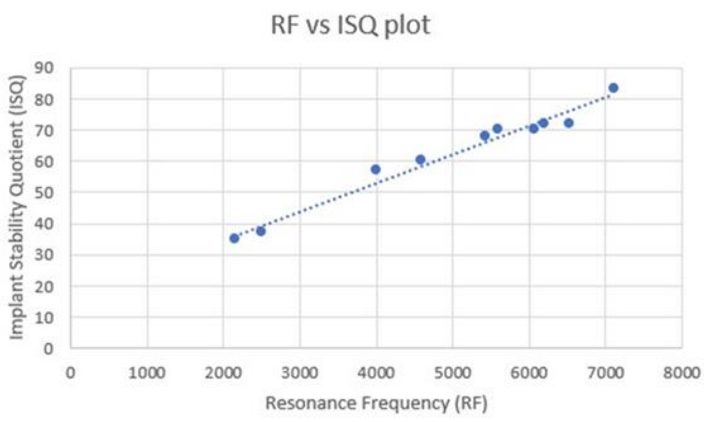

Figure 36: Resonance Frequency (RF) vs Implant Stability Quotient (ISQ) plot

It is clearly evident in the plot that the values have a purely increasing trend and based on the trendline, the increasing trend is almost completely linear. The regression statistics can be seen in Table 5.

Table 17: Regression Statistics

Regression Statistics	
Multiple R	0.986376419
R Square	0.972938441
Adjusted R Square	0.969555746
Standard Error	2.719544851
Observations	10

The R squared value represents the proportion of variance for the dependent variable which can be understood from the independent variable whereas the adjusted R squared value is the value which is adjusted based on the number of predictors present. It increases when the predictors improve the model more than expected whereas it decreases when the predictors improve the model less than expected. In the experimentation conducted, the R square value of 97.2% is achieved whereas the adjusted R square value of 97% is achieved. Moreover, we also performed ANOVA analysis and the results obtained are displays in Table 6 and 7

Table 18: ANOVA Analysis-I

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	2127.232606	2127.233	287.6223	1.4826E-07
Residual	8	59.16739357	7.395924		
Total	9	2186.4			

Table 19: ANOVA Analysis -II

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	16.46801861	2.841605651	5.795322	0.000407
X Variable 1	0.009147428	0.000539371	16.95943	1.48E-07

The most important parameter to be observed in ANOVA is the p value of the analysis. The p value of any experimentation is based upon two hypotheses; null hypothesis and alternative hypothesis. The null hypothesis denotes those whatever differences may be observed in analysis are not real and the variation in the dataset is only because of random variations. The alternative hypothesis claims that the null hypothesis is untrue. The probability of observing deviations from the null hypothesis given that the null hypothesis is correct is termed as the p value of the analysis. If the p value is lesser than a specific number (in most of the cases lesser than 0.05), then the result is considered statistically significant and the alternative hypothesis is considered as true and the null hypothesis is rejected. In the experimentation conducted, the resultant p value was 0.000407, which is lesser than 0.05. This value proves that as the RF increases, the ISQ also increases thus proving the strong linear relationship between the two variables. The equation of correlation between the two variables is mentioned in equation 3.

$$ISQ = 0.0091*(RFA) + 16.468 \quad (3)$$

Conclusion

This project focused to find the relationship between the resonance frequency and the implant stability quotient (ISQ). The Resonance Frequency Analysis technique for primary stability measurement was the focal point of the research. Mathematical, CAD, Numerical models were developed to study the relationship between resonance frequency and Implant stability quotient. A CAD geometry was developed and modal analysis was performed to analyze the natural frequency and different mode shapes for the measurement of the implant stability quotient. Further on, vibro-acoustic base experimental setup was developed to test and verify the results and determine the relationship between resonance frequency and implant stability quotient.

The following conclusion can be drawn from the research conducted in the project

1. Among the various RFA techniques the electromagnetic RFA technique is the most popular method with some of the available commercial devices
2. Vibro-acoustic is another emerging technique which can be used to measure the primary stability of the dental implant. However, further research is needed in-order to have accurate results
3. From Numerical analysis the results obtained showed that the natural frequency of the dental implant systems lies between 3KHz to 9KHz with different mode shapes obtained at various frequencies
4. Variation in input signal after 10KHz showed large variations in the Resonance frequency and ISQ value
5. The results obtained from the experimentation as well as from Modal analysis were studied and a co-relation between RFA and ISQ was successfully defined.

Future Scope

The main future scope of this project would be to miniaturize the experimental setup for the prototype development of the handheld device to measure the dental implant stability. Different parameters can be varied to understand the accuracy and reliability of the device. By comparing the results obtained from experimental setup to the existing commercial devices. Once the miniaturization is done the next step would be the manufacturing of the device. Its design will have to be modified to make it user-friendly and ergonomic device with a low cost of production so that it can be made accessible to a larger market.

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